Supplementary Information for

Physicians' artificial intelligence training needs in continuing medical education: a nationwide survey across adoption groups

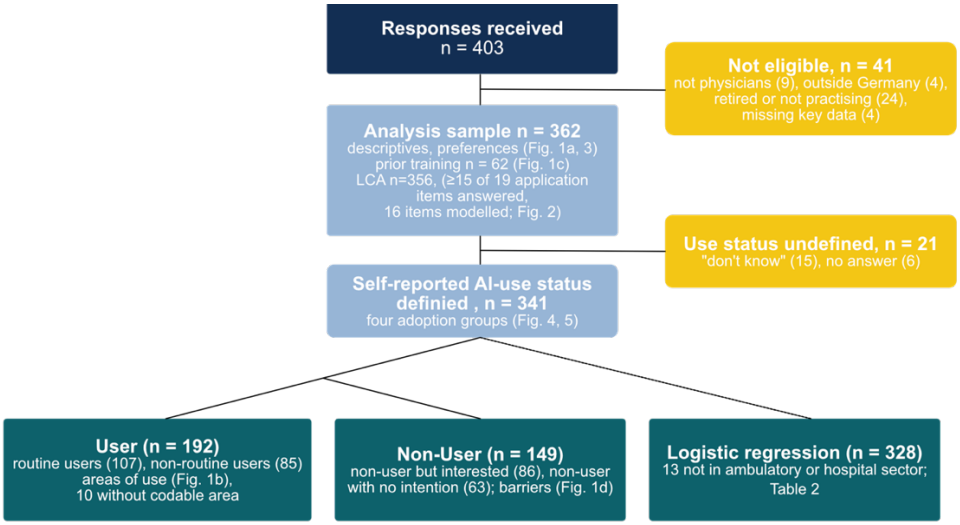
Supplementary Notes Overview

The Supplement for this paper contains the Supplementary Notes 1-10. Supplementary Figures and Supplementary Tables are numbered consecutively throughout the Supplementary Information:

Supplementary Note 1:	Supplementary Fig. 1 and Supplementary Table 1
Supplementary Note 2:	Supplementary Tables 2–5
Supplementary Note 3:	Supplementary Tables 6–9
Supplementary Note 4:	Supplementary Tables 10–14
Supplementary Note 5:	Supplementary Table 15
Supplementary Note 6:	Supplementary Fig. 2 and Supplementary Tables 16–18
(Supplementary Note 7:	the questionnaire contains no figures or tables)
Supplementary Note 8:	Supplementary Fig. 3 and Supplementary Tables 19–29
Supplementary Note 9:	Supplementary Table 30
Supplementary Note 10:	Supplementary Tables 31–44.

All numbers were generated by the scripts named at the end of each note. The scripts of the submitted analysis are in a public code repository, those added in revision in its folder revision/.

Supplementary Note 1: Participant flow and exclusion rules



Supplementary Fig. 1 | STROBE-style participant flow diagram. Of 403 responses received, 41 were not eligible (not physicians, n = 9; practising outside Germany, n = 4; retired or not in active clinical practice, n = 24; missing key demographic data, n = 4), yielding an analysis sample of 362 physicians used for descriptive analyses and training preferences (Fig. 1a and Fig. 3); 62 of these had received prior AI-specific training (Fig. 1c), and 356 with at least 15 of the 19 application items answered entered the latent class analysis (16 items modelled; Fig. 2). Self-reported AI-use status was defined for 341 respondents (undefined for 21: 'don't know', n = 15; no answer, n = 6), who form the four adoption groups (Figs. 4 and 5): 192 users (107 routine, 85 non-routine; areas of use in Fig. 1b, ten without a codable area) and 149 non-users (86 interested, 63 without current intention; barriers in Fig. 1d). The multivariable logistic regression (Table 2) comprised 328 respondents after excluding 13 not working in the ambulatory or hospital sector.

Data cleaning was performed on the raw sample of n = 403 responses. No duplicate entries were identified. Respondents were excluded stepwise against predefined eligibility criteria (Supplementary Table 1): respondents who were not physicians (other healthcare professionals, nursing staff, medical assistants, or medical students; n = 9; free-text and inconsistently spelled job titles were standardised before this step); respondents practising outside Germany (Austria 1, Switzerland 3; n = 4); respondents who were retired or not working in a clinical environment (retirement, self-governance or politics, administration or management, and individual cases indicating no clinical practice; n = 24) and respondents with missing values in key demographic variables (federal state 2, age 1, sex 1; n = 4). This yielded the final analytic sample of n = 362, comprising physicians from all 16 federal states. Of the 24 respondents in the third group, seven had retired, five worked in self-governance or politics, four in administration or management, and eight were individual cases, among them one occupational physician who gave no work setting (#284) and whose remaining answers indicated no clinical workplace. The companion analysis of unsanctioned AI use in the same survey (ref. 22 of the main

text) retained this record and therefore starts from a cleaned sample of 363 before its own analytical exclusions; the two samples otherwise coincide.

In a second pass, a logical-consistency check of the employer-provision items identified three records in which respondents reported ‘no tools available’ or use of freely available AI while simultaneously indicating a form of employer provision (#89, #345, #316). These records were recoded to ‘no employer provision’ and the corresponding ‘form of provision’ field was cleared, following predefined rules (codebook, Supplementary Note 4). They were corrected, not excluded, and remain in the analytic sample.

Supplementary Table 1 | Reasons for ineligibility

Reason for ineligibility	n	Source column in the raw export
Not a physician (other occupation, medical assistant, nurse or student)	9	M
Working outside Germany (Austria 1, Switzerland 3)	4	J
Retired (7) or not working in a clinical environment (politics 5, administration 4, individual cases 8)	24	J, M, L, AN, BE
Missing values in key demographic variables (federal state 2, age 1, sex 1)	4	J, I, K

Script: prepare_data.py

Supplementary Note 2: Participant characteristics and comparison with the physician population

Supplementary Table 2 | Full distribution of age, sex, work setting, professional experience and self-rated digital competence (n=362), Values are n (%).

Characteristic	n (%)
Age group	
20–29 years	27 (7.5)
30–39 years	104 (28.7)
40–49 years	60 (16.6)
50–59 years	86 (23.8)
60–69 years	66 (18.2)
≥70 years	19 (5.2)
Sex	
Female	183 (50.6)
Male	179 (49.4)
Work setting	
Ambulatory care – solo practice	77 (21.3)
Ambulatory care – group practice	85 (23.5)
Ambulatory care – medical care centre (MVZ)	33 (9.1)
Hospital – inpatient	136 (37.6)
Hospital – outpatient	16 (4.4)
Public health service	5 (1.4)
Emergency medical service	5 (1.4)
No information	5 (1.4)
Professional experience (since licensure)	
0–4 years	65 (18.0)
5–9 years	68 (18.8)
10–14 years	33 (9.1)
15–19 years	19 (5.2)
20–25 years	30 (8.3)
>25 years	145 (40.1)
No information	2 (0.6)
Self-rated professional digital competence	
1 (very low)	11 (3.0)
2	65 (18.0)
3	141 (39.0)
4	100 (27.6)
5 (very high)	45 (12.4)
Mean ± SD	3.28 ± 1.00

Professional experience was assessed from the third state examination (licensure). One respondent reported experience only in the item for non-physician staff (counted under >25 years); 'no information' refers to explicit non-response (n = 2). Digital competence was self-rated in the professional context on a 5-point scale (1 = very low, 5 = very high).

Supplementary Table 3 | Federal state of practice (n = 362). Values are n (%).

Characteristic	n (%)
North Rhine-Westphalia	65 (18.0)
Bavaria	62 (17.1)

Baden-Württemberg	57 (15.7)
Lower Saxony	32 (8.8)
Saxony	25 (6.9)
Rhineland-Palatinate	22 (6.1)
Berlin	21 (5.8)
Hesse	20 (5.5)
Schleswig-Holstein	15 (4.1)
Brandenburg	13 (3.6)
Hamburg	11 (3.0)
Mecklenburg-Western Pomerania	5 (1.4)
Saxony-Anhalt	5 (1.4)
Thuringia	5 (1.4)
Saarland	3 (0.8)
Bremen	1 (0.3)

Supplementary Table 4 | Medical specialty (n = 362). Values are n (%). Categories as reported by participants.

Characteristic	**n (%)**
General practice	70 (19.3)
Internal medicine	41 (11.3)
Paediatrics	27 (7.5)
Gynaecology and obstetrics	27 (7.5)
Orthopaedics and trauma surgery (incl. plastic/reconstructive surgery)	26 (7.2)
Psychiatry	21 (5.8)
Anaesthesiology	17 (4.7)
Neurology	13 (3.6)
Ophthalmology	12 (3.3)
Cardiology	11 (3.0)
Radiology	11 (3.0)
Dermatology	9 (2.5)
General surgery	9 (2.5)
Psychosomatic medicine	8 (2.2)
Otorhinolaryngology	7 (1.9)
Child and adolescent psychiatry	6 (1.7)
Occupational medicine	5 (1.4)
Cardiothoracic and vascular surgery	4 (1.1)
Urology	4 (1.1)
Radiation therapy	3 (0.8)
Geriatrics	3 (0.8)
Laboratory medicine	3 (0.8)
Phoniatrics and paediatric audiology	2 (0.6)
Oral and maxillofacial surgery	2 (0.6)
Neurosurgery	2 (0.6)
Pathology	2 (0.6)
Gastroenterology	2 (0.6)
Palliative medicine	2 (0.6)
Transfusion medicine	2 (0.6)
No information	2 (0.6)
Microbiology, virology and infection epidemiology	1 (0.3)
Rehabilitation medicine	1 (0.3)
Nephrology	1 (0.3)

Orthopaedics	1 (0.3)
Physical and rehabilitative medicine	1 (0.3)
Public health	1 (0.3)
Intensive care medicine	1 (0.3)
Dentistry	1 (0.3)
Others	1 (0.3)

Supplementary Table 5 | Comparison of the study sample with the German physician population (Bundesärztekammer physician statistics, 31 December 2025).

Characteristic	**Study sample (N = 362), %**	**German physicians (BÄK 2025, N = 446,120), %**
Age group		
<40 years	36.2	33.1
40–49 years	16.6	23.3
50–59 years	23.8	20.2
≥60 years	23.5	23.4
Sex		
Female	50.6	50.5
Male	49.4	49.5
Care setting		
Ambulatory	53.9	39.0
Hospital (inpatient and outpatient)	42.0	52.3
Other / no information	4.1	8.7

BÄK reference population: all professionally active physicians in Germany (N = 446,120). BÄK age groups (≤34, 35–39, 40–49, 50–59, 60–65, >65 years) were collapsed to match the survey categories. BÄK care settings: ambulatory (n = 173,900), inpatient (n = 233,500), public authorities/corporations and other areas (n = 38,700). Survey 'ambulatory' comprises solo practice, group practice and MVZ; survey 'hospital' comprises inpatient and hospital-based outpatient care. Source: Bundesärztekammer, Ärztestatistik zum 31. Dezember 2025

Script: prepare_data.py / results_R2.py

Supplementary Note 3: Multivariate logistic regression of professional AI use

Scope of Note 3

The regression reported in Table 2 contains private AI use as a characteristic of the physician, because without any measure of personal engagement with AI the conclusion that no individual characteristic is associated with use would be conditional on the characteristics entered. It contains no composite index of digital application use; Supplementary Table 7 documents why: such an index combines availability, familiarity and use in a single score, and its association with AI use arises from applications that are themselves AI-based. Supplementary Table 6 reports the robustness of the model in Table 2 to coding, sample and the addition of summaries of the application items; Supplementary Table 7 examines the composite index within the covariate set without private AI use; Supplementary Table 8 defines the variables; Supplementary Table 9 gives the item-level analysis. All models are logistic regressions on complete cases with Wald confidence intervals; block tests are likelihood-ratio tests.

Supplementary Table 6 | Robustness of the model in Table 2

Analysis	n	Parameters	Employer provision: yes, OR (95% CI)	Private AI use, OR (95% CI)	Digital competence, OR (95% CI)	Other terms
Main model (Table 2)	328	10	14.39 (6.53-31.72)	4.11 (2.34-7.21)	1.34 (1.01-1.77)	
Without private AI use	328	9	13.47 (6.25-29.04)	–	1.39 (1.06-1.82)	
Employer provision binary ('don't know' recoded to 'no')	328	9	14.40 (6.60-31.45)	4.11 (2.35-7.20)	1.34 (1.01-1.77)	
Excluding solo practices	256	10	17.38 (6.56-46.05)	3.15 (1.66-5.97)	1.59 (1.14-2.20)	
Sector in three categories (hospital / MVZ / practice)	328	11	14.30 (6.48-31.54)	4.08 (2.32-7.16)	1.34 (1.02-1.78)	MVZ (ref. hospital): 1.65 (0.58-4.70), p = 0.347
Age in five categories (20-29 separate)	328	11	14.17 (6.44-31.20)	4.01 (2.28-7.06)	1.36 (1.03-1.81)	Age 20-29 (ref. 30-39): 3.34 (1.08-10.38), p = 0.037
Private engagement with AI in three categories instead of private use	328	11	14.39 (6.53-31.71)	–	1.34 (1.01-1.78)	Private engagement with AI: informs only (ref. hardly at all): 1.05 (0.36-3.03), p = 0.927
Employed physicians only (hospital inpatient and outpatient, MVZ)	176	10	16.93 (5.33-53.80)	4.29 (1.84-10.00)	1.74 (1.16-2.59)	MVZ (ref. hospital): 1.36 (0.43-4.33), p = 0.598
Practice-based physicians only (solo and group practices)	152	9	15.44 (4.89-48.71)	4.23 (1.85-9.64)	0.99 (0.64-1.55)	
Hospital physicians only	144	9	16.99 (4.79-60.22)	3.14 (1.28-7.70)	1.71 (1.09-2.69)	
Main model plus the three 16-item counts	322	13	13.10 (5.85-29.33)	4.03 (2.27-7.15)	1.26 (0.93-1.69)	Applications not available (per application, 0-16): 0.94 (0.85-1.04), p = 0.244
Main model plus the 16-item composite score (per SD)	322	11	12.27 (5.51-27.32)	4.18 (2.36-7.40)	1.25 (0.93-1.67)	Composite score, 16 items (per SD): 1.43 (1.03-1.97), p = 0.030
Main model plus the counts on 14 items without AI-based applications	322	13	13.58 (6.07-30.35)	4.02 (2.27-7.13)	1.29 (0.96-1.74)	Applications not available (per application, 0-14): 0.96 (0.86-1.07), p = 0.476

Employed physicians: hospital inpatient, hospital outpatient and medical care centre (MVZ) respondents, for whom employer and respondent are organisationally distinct; ownership status was not surveyed, so practice-based respondents include employed physicians in group practices. In the main model with sector in three categories, the interaction between provision and practice-based setting was not significant (likelihood-ratio $\chi^2(2) = 1.17$, $p = 0.56$; provision OR 16.1 in hospitals and MVZ, 13.0 in practices). AI use with employer provision was 90.5% in hospitals (n = 42), 91.7% in MVZ (n = 12) and 89.4% in practices (n = 47); without provision 42.5%, 56.2% and 39.1%.

Composite index of digital application use

A composite index was defined as the mean coded use frequency of the 19 application items (daily 4, weekly 3, monthly 2, rarely 1), with 'I would like to, but it is not available' and 'unclear what this is' both scored 0, computed for respondents with at least 15 valid items and entered per standard deviation. In the covariate set without private AI use it is associated with AI use (OR 1.48 per SD, 95% CI 1.09–2.00; Supplementary Table 7, first row).

The objection that such an index combines availability, familiarity and use was examined in four ways: First, the index was replaced by its components on the 16 items retained in the latent class analysis (Supplementary Note 8): the number of applications reported as not available, the number not recognised, and the mean use frequency among the remaining applications. Only the number not recognised was associated with AI use (OR 0.84 per application), and the association was attenuated once private AI use was added (OR 0.87, 0.74–1.01).

Second, 'unclear what this is' was treated as missing rather than as zero use; the index was then no longer associated with AI use (OR 1.18, 0.89–1.58; with at least 15 recognised items required, n = 261, OR 1.25, 0.91–1.73).

Third, availability alone, the share of answered applications not reported as unavailable, was not associated with AI use (OR 0.98 per SD).

Fourth, because the surveyed applications include AI-based ones, the index was recomputed without clinical decision support and speech recognition (14 items) and additionally without phone assistance systems and automated duty scheduling (12 items). Its association with AI use then disappeared (OR 1.26, 0.94–1.70, and 1.15, 0.87–1.54). At the level of single items (Supplementary Table 9), weekly or daily use of clinical decision support (n = 42; OR 4.67, 1.51–14.44), of phone assistance systems (n = 51; OR 4.11, 1.55–10.86) and of video consultation (n = 41; OR 3.11, 1.16–8.36) was associated with professional AI use within the same covariate set. No other item was.

The composite index therefore measured, in part, the outcome itself, and its remaining association reflected unfamiliarity with the surveyed applications rather than any property of the work environment. It is not retained as a predictor; the number of applications not recognised is reported descriptively in relation to the use profiles (Supplementary Note 8, Supplementary Table 29; Fig. 2d).

Supplementary Table 7 | The composite index examined (covariate set without private AI use: age, sex, digital competence, prior training, sector, employer provision; n = 322 unless stated)

Analysis	Index or components: OR (95% CI), p	Employer provision: yes, OR (95% CI)
Index on 19 items, per SD	1.48 (1.09-2.00), p = 0.012	11.38 (5.24-24.74)
Index replaced by its components (16 items): not available, not recognised, use frequency among the rest	Applications not available (per application, 0-16): 0.96 (0.88-1.06), p = 0.461 · Applications not recognised (per application, 0-16): 0.84 (0.73-0.97), p = 0.021 · Mean use frequency among the remaining applications (per point, 1-4): 1.27 (0.79-2.04), p = 0.328	13.03 (5.92-28.69)
Index with 'unclear what this is' treated as missing (19 items;	Index with 'unclear' as missing, 19 items (per SD): 1.18 (0.89-1.58), p = 0.255	12.31 (5.65-26.83)

Analysis	Index or components: OR (95% CI), p	Employer provision: yes, OR (95% CI)
denominator = answered, recognised items)		
Availability index: share of answered applications not reported as unavailable (19 items), per SD	Availability index, 19 items (per SD): 0.98 (0.75-1.27), p = 0.878	13.49 (6.21-29.31)
Index on the 16 items of the latent class analysis, per SD	Composite score, 16 items (per SD): 1.39 (1.03-1.88), p = 0.032	11.66 (5.37-25.34)
Index on 14 items without AI-based applications (clinical decision support, speech recognition), per SD	Composite score, 14 items without AI-based applications (per SD): 1.26 (0.94-1.70), p = 0.126	12.19 (5.62-26.44)
Index on 12 items (additionally without phone assistance, automated duty scheduling), per SD	Composite score, 12 items (per SD): 1.15 (0.87-1.54), p = 0.328	12.75 (5.89-27.60)
Components on the 14 items without AI-based applications	Applications not available (per application, 0-14): 0.99 (0.89-1.10), p = 0.811 · Applications not recognised (per application, 0-14): 0.86 (0.73-1.01), p = 0.068 · Mean use frequency among the remaining applications, 14 items (per point, 1-4): 1.12 (0.71-1.77), p = 0.629	13.44 (6.12-29.52)
Sector-neutral index (8 items; submitted sensitivity)	Sector-neutral index, 8 items (per SD): 1.45 (1.09-1.93), p = 0.011	11.83 (5.44-25.71)
Components (16 items) with private AI use added	Applications not available (per application, 0-16): 0.94 (0.85-1.04), p = 0.244 · Applications not recognised (per application, 0-16): 0.87 (0.74-1.01), p = 0.069 · Mean use frequency among the remaining applications (per point, 1-4): 1.28 (0.78-2.13), p = 0.331	13.10 (5.85-29.33)

Supplementary Table 8 | Variables entered into the multivariable logistic regression (Table 2) and its sensitivity analyses, definitions and coding

Variable	Definition	Coding
Professional AI use (outcome)	Use of AI applications in professional routine	1 = regularly, occasionally or rarely; 0 = no; 'don't know' and no answer excluded (n = 341 with defined status)
Age	Four categories	<40 (ref.), 40–49, 50–59, ≥60 years
Sex	Self-reported	male vs female (ref.)
Self-rated professional digital competence	Single item, 1 = very low to 5 = very high	linear, per point (categorical coding did not improve fit: $\chi^2(3) = 1.54$, p = 0.67)
Prior AI-specific training	Training specifically on AI received in a professional context	yes vs no (ref.)
Private AI use	'I use AI applications privately (e.g. ChatGPT, AI image generators)' selected in the multiple-response item on private engagement with AI	yes vs no (ref.)
Sector	Main work setting	ambulatory (solo practice, group practice, MVZ) vs hospital (inpatient, outpatient; ref.); public health service, emergency medical service and no answer excluded
Employer provision of AI applications	'Does your employer provide you with AI-based applications?'	yes, don't know, vs no (ref.)
Application counts (sensitivity analyses only)	16 application items retained in the latent class analysis	number reported as not available (0–16); number not recognised (0–16); mean use frequency among the remaining applications (1 = rarely to 4 = daily); respondents with at least 15 valid items of 19

Supplementary Table 9 | Weekly or daily use of each application item added to the covariate set without private AI use (n = 322; the item's 'unclear what this is' and 'not available' indicators entered alongside)

Item	n weekly–daily	Weekly–daily use: OR (95% CI), p
Practice/hospital IS	300	1.51 (0.38-6.03), p = 0.563
Electronic patient record (ePA)	154	0.67 (0.33-1.35), p = 0.263
e-medication list	158	0.83 (0.44-1.60), p = 0.585
e-sick note (eAU)	181	0.51 (0.25-1.05), p = 0.067
PACS / imaging	183	1.51 (0.77-2.96), p = 0.228
Laboratory IS	239	0.79 (0.35-1.79), p = 0.573
Clinical decision support	42	4.67 (1.51-14.44), p = 0.007
DiGA	56	1.49 (0.69-3.20), p = 0.307
Health apps	57	1.20 (0.57-2.52), p = 0.628
Phone assistance	51	4.11 (1.55-10.86), p = 0.004
Video consultation	41	3.11 (1.16-8.36), p = 0.025
Tele-/home monitoring	31	1.97 (0.67-5.80), p = 0.221
Speech recognition	97	1.15 (0.57-2.34), p = 0.690
Medication safety (AMTS)	124	1.36 (0.70-2.64), p = 0.360
Automated duty scheduling	23	2.89 (0.71-11.79), p = 0.139
Knowledge management	204	1.61 (0.86-2.99), p = 0.134

Script: analysis/regression.py (Table 2; Supplementary Table 6, 7, 9)

Supplementary Note 4: Codebook: free-text recoding, variable definitions and correction rules

Free-text recoding of open-ended responses

Three multiple-response items offered an open-text option in addition to predefined categories: areas of AI use (Fig. 1b), formats of prior AI-specific training (Fig. 1c), and barriers to AI use (Fig. 1d). Predefined options were retained verbatim; free-text entries were mapped to an existing category where semantically equivalent, or marked uncodable (–). The authoritative mappings are implemented in analysis/fig1.py, which produces Fig. 1 (results_R2.py in the repository root contains the same mappings except the entry 'RECHERCHE' noted below).

Supplementary Table 10 | Areas of AI use – recoding of free-text entries

Item: Q20 'Für welche der folgenden Bereiche nutzen Sie KI-Anwendungen? (Mehrfachnennungen möglich)' ('For which of the following areas do you use AI applications? (Multiple answers possible)'; users and respondents unsure of their use). Predefined categories (German original in brackets): Administration [Administrative Aufgaben], Diagnostics [Diagnostik und Befund], Treatment planning [Behandlungsplanung und Therapieentscheidungen], Research [Forschung], Patient communication [Patientenkommunikation und Service].

Free-text entry (German)	Assigned category
Recherche	Research
RECHERCHE	Research
Recherche, Patienteninformation	Research + Patient communication
Übersetzungen für fremdsprachliche Patienten	Patient communication
Spracherkennung und Literaturrecherche	Research + Administration
Behandlungsdokumentation	Administration
OpenEvidence	Research
Diktiergerät	Administration
Marketing	Administration
Dokumentation / Spracherkennung	Administration
Unterstützung bei seltenen Diagnosen …	Diagnostics
Bei der Strahlentherapie selbst	Diagnostics
zusätzliche Auswertung der Lunge	Diagnostics
Erstellen von Prüfungsfragen	– (uncodable)
Ich benutze meinen privaten Account	– (uncodable)
Nur wenig Nützliches … für Dienstplan etc. …	– (uncodable)
Exposition	– (uncodable)
nur mal so zum Nachschlagen und bestätigen …	– (uncodable)

Ten of the 192 users contributed no codable area (six selected 'no answer'; four gave uncodable free text). They remain in the denominator (n = 192). The entry 'RECHERCHE' (upper case) is mapped to research in the same way as 'Recherche'; this mapping is implemented in analysis/fig1.py, which produces Fig. 1.

Supplementary Table 11 | Formats of prior AI-specific training – recoding

Item: Q23 ‘Welche Art von KI-Schulung war das? (Mehrfachnennung möglich)’ (‘What type of AI training was it? (Multiple answers possible)’; trained physicians, n = 62).

Response (German)	Category
Kurzvortrag/Informationsveranstaltung (<2 Stunden)	Short lecture (<2 h)
Online-Modul/E-learning	E-learning module
Workshop (wenige Stunden bis 1 Tag)	Workshop
Integrierte in anderen Fortbildungen	Integrated in other CME
Mehrtägiger Kurs/Zertifikatslehrgang	Certificate / multi-day
„Seminar des Hartmannbund … nach 15 min abgebrochen”	Short lecture (<2 h)
Keine Antwort	– (uncodable)

Supplementary Table 12 | Barriers to AI use – recoding

Item: Q16 ‘Was sind die Hauptgründe, warum Sie derzeit keine KI-basierten Anwendungen nutzen?’ (‘What are the main reasons why you currently do not use AI-based applications?’; at most three selections; analysed for definite non-users, n = 149).

Response / free-text entry (German)	Category
Nicht verfügbar	Not available
Fehlendes Wissen/Schulungen	Lack of knowledge / training
Sicherheits-/Datenschutzbedenken	Data protection / security concerns
Mangelnde Integrationsmöglichkeiten in bestehenden Systemen	Insufficient system integration
Keine wahrgenommene Nützlichkeit/Zeitgewinn	No perceived benefit
fehlende Kapazitäten / Zeitmangel für Implementierung	Lack of time / capacity
kein bedarf	No perceived benefit
„Als sporadisch in der Notfallpraxis tätiger Arzt … wird auch nicht bereitgestellt”	Not available
„Die Firmen … haben KI noch nicht … implementiert”	Not available
Kosten / „Kh hat kein Geld”	Costs
Testphase	Other

Costs and Other were each named by fewer than five respondents and are therefore not shown in Fig. 1d (see its caption).

Variables and coding for the regression, the application counts and the latent class analysis

Variables entered into the multivariable logistic regression (Table 2) and its sensitivity analyses (Supplementary Note 3), into the comparisons across adoption groups (Fig. 4; Supplementary Notes 6 and 10) and into the latent class analysis (Fig. 2; Supplementary Note 8) were derived from the questionnaire as follows.

Supplementary Table 13 | Variable definitions and coding

Variable	Questionnaire item (English translation; German original in Supplementary Note 7)	Coding
AI use (outcome)	Q19 ‘Do you already use AI-based applications in your professional daily routine?’	1 = any affirmative answer (‘regularly’, ‘occasionally’, ‘rarely / experimentally’); 0 = ‘No, but planned / interested’ or ‘No, and not currently planned’; ‘I don’t know / am unsure’ and no answer = missing (excluded from analyses requiring a defined use status; n = 341)

Variable	Questionnaire item (English translation; German original in Supplementary Note 7)	Coding
Adoption group	Q19 (same item)	routine use ('Yes, regularly (several times per week)', n = 107) · non-routine use ('Yes, occasionally (once per week)' or 'Yes, rarely / experimentally (once per month)', n = 85) · interested non-user ('No, but planned / interested', n = 86) · no current intention ('No, and not currently planned', n = 63)
Age (age4)	Q3 'How old are you?'	<40 [ref] (20–29, 30–39) · 40–49 · 50–59 · ≥60 (60–69, ≥70)
Sex (male)	Q5 'Which gender do you identify with?'	1 = male; 0 = female [ref]
Digital competence	Q17 'How would you rate your digital competence in a professional context?'	continuous, 1–5
Prior AI training (trained)	Q22 'Have you already received training specifically on the topic of AI in a professional context?'	1 = yes; 0 = no [ref]
Sector (ambulatory)	Q6 'In which setting do you mainly work?'	1 = ambulatory (single practice, group practice, MVZ); 0 = hospital [ref] (inpatient, outpatient); other settings and no answer = missing
Employer provision	Q14 'Does your employer provide you with AI-based applications?'	yes · no [ref] · don't know
Private AI use (private_ai)	Q21 'How do you engage with the topic of AI in your private life? (Multiple answers possible)'	1 = 'I use AI applications privately (e.g. ChatGPT, AI image generators)' selected; 0 = not selected [ref]
Professional experience (early career)	Q9 'How much professional experience do you have? (Counted from the third state examination)'	0–4 years (n = 65) versus 5 years or more (n = 295); 2 without information
Unauthorised use (exposure marker, Supplementary Note 10 only)	Q32 'Are you aware of cases of Shadow AI in your professional environment?'	1 = 'Yes, I have used AI myself without formal authorisation before'; 0 = other substantive answer; no answer = missing
LCA indicators	Q13, 16 application items (Supplementary Table 14; items 12, 15 and 16 excluded)	four states per item: not available ('I would like to, but it is not available'; the 20 unanswered cells of 16 respondents, 0.4% of all cells, were also coded as not available) / not recognised ('unclear what this is') / rarely to monthly / weekly to daily; respondents with ≥15 of 19 items answered (n = 356). Sensitivity analyses in Supplementary Note 8.
Application counts (Supplementary Note 3; Fig. 2d)	Q13, 16 application items (Supplementary Table 14)	number reported as 'not available' (0–16); number reported as 'unclear what this is' (0–16); mean frequency among the remaining items (rarely = 1 … daily = 4); ≥15 valid items of 19 (n = 356)
Composite index (Supplementary Note 3 only)	Q13, 19 application items (Supplementary Table 14)	mean of item frequencies (0–4) with both non-use states scored 0; ≥15 valid items required; z-standardised within the analysis sample

Frequency coding of the application items for Fig. 2a: daily = 4; weekly = 3; monthly = 2; rarely = 1; 'I would like to, but it is not available' and 'unclear what this is' are not use states and are displayed as 0.

Supplementary Table 14 | The 19 digital-application items (* = sector-neutral subset used in Supplementary Table 7; items 12, 15 and 16 are excluded from the latent class analysis and the application counts)

No.	Item (German)	Short label (Fig. 2a)	Neutral
1	Praxisverwaltungssysteme (PVS) / Krankenhausinformationssystem (KIS)	Practice/hospital IS	*
2	Elektronische Patientenakte (ePA)	Electronic patient record (ePA)	
3	Elektronische Medikationsliste (eML/eMP)	e-medication list	*
4	Elektronische Arbeitsunfähigkeit (eAU)	e-sick note (eAU)	
5	Bildbetrachtungssysteme (z.B. PACS)	PACS / imaging	

No.	Item (German)	Short label (Fig. 2a)	Neutral
6	Laborinformationssysteme (LIS)	Laboratory IS	*
7	Differentialdiagnostik bzw. Clinical-Decision-Support (CDSS)	Clinical decision support	*
8	Digitale Gesundheitsanwendungen (DiGA)	DiGA	
9	Gesundheits-Apps für die Patientenversorgung	Health apps	
10	Telefon-Assistenzsysteme (z.B. Terminvergabe)	Phone assistance	
11	Videosprechstunde	Video consultation	
12	Tele-Visite (z.B. Intensivstation / Stroke-Unit / Pflegeheim)	Tele-rounds	*
13	Tele-/Home-Monitoring	Tele-/home monitoring	
14	Spracherkennung zur automatisierten Behandlungs-Dokumentation	Speech recognition	
15	Digitale Patientenvoranmeldung (NIDA, IVENA, Pulsation)	Digital pre-registration	
16	Ersteinschätzungs-Tools (docyet, ada, SMeD)	Triage tools	*
17	Arzneimittel-Therapiesicherheitssystem (AMTS)	Medication safety (AMTS)	*
18	Automatisierte Dienstplanung	Automated duty scheduling	*
19	Wissensmanagement (z.B. Amboss, Elsevier, Thieme)	Knowledge management	

Correction rules for logically inconsistent records

During data cleaning (Supplementary Note 1), a logical-consistency check was applied to the employer-provision items. Records in which a respondent indicated that their employer provides AI applications (or named a form of provision) while simultaneously reporting that no tools were available, or that they used only freely available ('shadow') AI, were treated as inconsistent.

Predefined rule: employer provision was recoded to 'no' and the associated 'form of provision' field was cleared. The rule was applied to three records (#89, #345, #316); all three remain in the analytic sample (n = 362).

This correction is distinct from the exclusion of records (Supplementary Note 1). One further record (#384) was excluded as a retired respondent and is not part of this correction.

Scripts: prepare_data.py, results_R2.py; analysis/fig1.py

Supplementary Note 5: Preferred format by age group (RQ3)

Supplementary Table 15 | Preferred formats among physicians under 40 and 50 years or older (n named / n in group).

Format	Under 40 years (n = 131)	50 years or older (n = 171)
Self-paced online course	102/131 = 77.9%	112/171 = 65.5%
Live webinars (interactive)	65/131 = 49.6%	87/171 = 50.9%
Information nuggets (15-30 min)	59/131 = 45.0%	50/171 = 29.2%
Blended learning	54/131 = 41.2%	56/171 = 32.7%
On-site sessions / workshops	47/131 = 35.9%	62/171 = 36.3%

Script: analysis/age_by_format.py (reproduces results_R4.py).

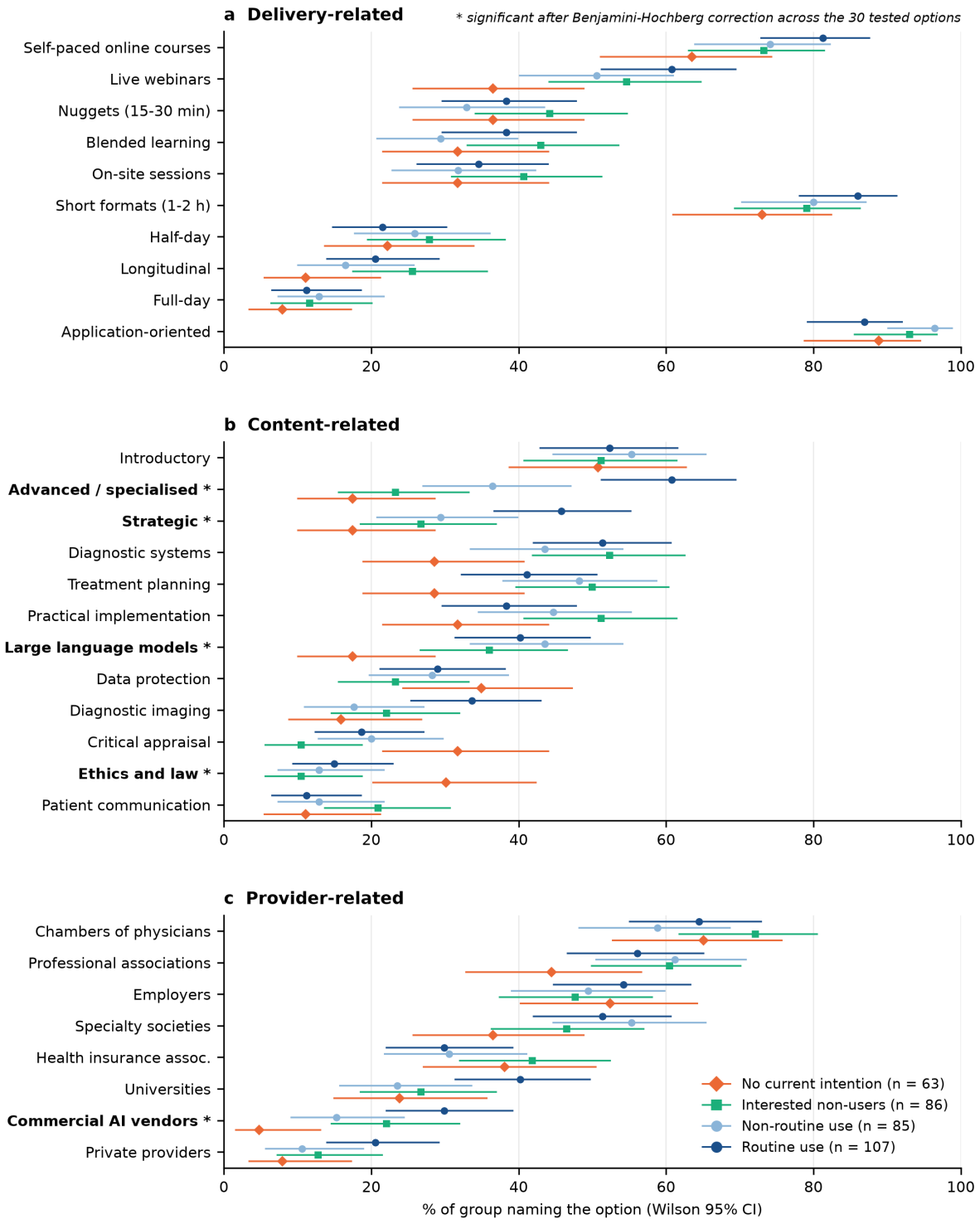
Supplementary Note 6: Training preferences across four adoption groups: omnibus tests, pairwise contrasts and equivalence bounds

Groups are defined from the item on professional AI use: non-users without current intention (n = 63), interested non-users (n = 86), non-routine users (occasional use, about once a week, or rare or experimental use, about once a month; n = 85) and routine users (regular use, several times a week; n = 107). 21 respondents who were unsure or gave no answer are excluded.

Supplementary Table 16 reports, for all 32 options, the percentage of each group naming the option with Wilson 95% confidence intervals, the four-group χ^2 statistic, Cramér's V, the raw p-value and the Benjamini–Hochberg-adjusted p-value across the 30 tested options; two options with expected cell counts below five were not tested.

Supplementary Table 17 reports pairwise Fisher's exact tests for the five omnibus-significant options across all six pairs, with risk differences and Newcombe 95% confidence intervals, corrected within the family of 30 pairwise tests.

Supplementary Table 18 reports, for the ten delivery-related options, the observed differences with 95% confidence intervals and the largest absolute difference not excluded at the 90% level (the smallest margin at which two groups would be declared equivalent in a two one-sided tests sense); the largest bound per option is Full-day 13, Application-oriented 16, Half-day 17, On-site sessions 21, Nuggets (15-30 min) 23, Short formats (1-2 h) 24, Longitudinal 24, Blended learning 25, Self-paced online courses 30, Live webinars 36 percentage points. Combining the two user categories was checked beforehand: rare and occasional users did not differ on any option after correction (smallest adjusted p = 0.40).



Supplementary Fig. 2 | All 30 tested preference options across the four adoption groups. Percentage of each group naming the option with Wilson 95% confidence intervals: non-users without current intention (orange diamonds, n = 63), interested non-users (green squares, n = 86), non-routine users (light blue circles, n = 85) and routine users (dark blue circles, n = 107); delivery-related (a), content-related (b) and provider-related (c) options, each ordered by descending overall endorsement. Asterisks and bold labels mark the five options significant after Benjamini–Hochberg correction across the 30 tests.

Supplementary Table 16 | All preference options across the four adoption groups: % (Wilson 95% CI), χ^2, Cramér's V, raw and BH-adjusted p. Asterisks mark the five options significant after correction.

Option	No current intention (n = 63)	Interested non-users (n = 86)	Non-routine use (n = 85)	Routine use (n = 107)	$\chi^2(3)$	Cramér's V	p	BH-adjusted p
Self-paced online courses	63.5 (51.1-74.3)	73.3 (63.1-81.5)	74.1 (63.9-82.2)	81.3 (72.9-87.6)	6.64	0.14	0.084	0.189
Live webinars	36.5 (25.7-48.9)	54.7 (44.2-64.7)	50.6 (40.2-61.0)	60.7 (51.3-69.5)	9.65	0.17	0.022	0.073
Nuggets (15-30 min)	36.5 (25.7-48.9)	44.2 (34.2-54.7)	32.9 (23.9-43.5)	38.3 (29.7-47.8)	2.38	0.08	0.498	0.597
Blended learning	31.7 (21.6-44.0)	43.0 (33.1-53.6)	29.4 (20.8-39.8)	38.3 (29.7-47.8)	4.18	0.11	0.242	0.346
On-site sessions	31.7 (21.6-44.0)	40.7 (30.9-51.3)	31.8 (22.8-42.3)	34.6 (26.2-44.0)	1.92	0.08	0.589	0.680
Short formats (1-2 h)	73.0 (61.0-82.4)	79.1 (69.3-86.3)	80.0 (70.3-87.1)	86.0 (78.2-91.3)	4.39	0.11	0.222	0.333
Half-day	22.2 (13.7-33.9)	27.9 (19.5-38.2)	25.9 (17.8-36.1)	21.5 (14.8-30.2)	1.33	0.06	0.723	0.803
Longitudinal	11.1 (5.5-21.2)	25.6 (17.5-35.7)	16.5 (10.1-25.8)	20.6 (14.0-29.2)	5.48	0.13	0.140	0.247
Full-day	7.9 (3.4-17.3)	11.6 (6.4-20.1)	12.9 (7.4-21.7)	11.2 (6.5-18.6)	0.95	0.05	0.813	0.841
Multi-day	1.6 (0.3-8.5)	7.0 (3.2-14.4)	3.5 (1.2-9.9)	5.6 (2.6-11.7)	-	-	-	not tested (expected count < 5)
Application-oriented	88.9 (78.8-94.5)	93.0 (85.6-96.8)	96.5 (90.1-98.8)	86.9 (79.2-92.0)	6.17	0.13	0.104	0.208
Introductory	50.8 (38.8-62.7)	51.2 (40.8-61.4)	55.3 (44.7-65.4)	52.3 (43.0-61.6)	0.40	0.03	0.940	0.940
Advanced / specialised *	17.5 (10.0-28.6)	23.3 (15.6-33.2)	36.5 (27.0-47.1)	60.7 (51.3-69.5)	43.06	0.36	< 0.001	< 0.001
Strategic *	17.5 (10.0-28.6)	26.7 (18.5-36.9)	29.4 (20.8-39.8)	45.8 (36.7-55.2)	16.91	0.22	< 0.001	0.007
Diagnostic systems	28.6 (18.9-40.7)	52.3 (41.9-62.6)	43.5 (33.5-54.1)	51.4 (42.0-60.7)	10.53	0.18	0.015	0.062
Treatment planning	28.6 (18.9-40.7)	50.0 (39.7-60.3)	48.2 (37.9-58.7)	41.1 (32.3-50.6)	8.18	0.15	0.042	0.116
Practical implementation	31.7 (21.6-44.0)	51.2 (40.8-61.4)	44.7 (34.6-55.3)	38.3 (29.7-47.8)	6.54	0.14	0.088	0.189
Large language models *	17.5 (10.0-28.6)	36.0 (26.7-46.6)	43.5 (33.5-54.1)	40.2 (31.4-49.7)	12.33	0.19	0.006	0.045
Data protection	34.9 (24.3-47.2)	23.3 (15.6-33.2)	28.2 (19.8-38.6)	29.0 (21.2-38.2)	2.45	0.08	0.484	0.597
Diagnostic imaging	15.9 (8.9-26.8)	22.1 (14.6-31.9)	17.6 (11.0-27.1)	33.6 (25.4-43.0)	9.89	0.17	0.020	0.073
Critical appraisal	31.7 (21.6-44.0)	10.5 (5.6-18.7)	20.0 (12.9-29.7)	18.7 (12.4-27.1)	10.60	0.18	0.014	0.062
Ethics and law *	30.2 (20.2-42.4)	10.5 (5.6-18.7)	12.9 (7.4-21.7)	15.0 (9.4-22.9)	11.95	0.19	0.008	0.045
Patient communication	11.1 (5.5-21.2)	20.9 (13.7-30.7)	12.9 (7.4-21.7)	11.2 (6.5-18.6)	4.61	0.12	0.202	0.320
AI in medical research	1.6 (0.3-8.5)	1.2 (0.2-6.3)	1.2 (0.2-6.4)	6.5 (3.2-12.9)	-	-	-	not tested (expected count < 5)
Chambers of physicians	65.1 (52.8-75.7)	72.1 (61.8-80.5)	58.8 (48.2-68.7)	64.5 (55.1-72.9)	3.34	0.10	0.342	0.446
Professional associations	44.4 (32.8-56.7)	60.5 (49.9-70.1)	61.2 (50.5-70.8)	56.1 (46.6-65.1)	5.03	0.12	0.170	0.283
Employers	52.4 (40.3-64.2)	47.7 (37.4-58.1)	49.4 (39.0-59.8)	54.2 (44.8-63.3)	0.95	0.05	0.812	0.841
Specialty societies	36.5 (25.7-48.9)	46.5 (36.3-57.0)	55.3 (44.7-65.4)	51.4 (42.0-60.7)	5.69	0.13	0.127	0.239
Assoc. of SHI physicians	38.1 (27.1-50.4)	41.9 (32.0-52.4)	30.6 (21.8-41.0)	29.9 (22.1-39.2)	3.99	0.11	0.263	0.358
Universities	23.8 (15.0-35.6)	26.7 (18.5-36.9)	23.5 (15.8-33.6)	40.2 (31.4-49.7)	8.61	0.16	0.035	0.105
Commercial AI vendors *	4.8 (1.6-13.1)	22.1 (14.6-31.9)	15.3 (9.2-24.4)	29.9 (22.1-39.2)	17.32	0.23	< 0.001	0.007
Private providers	7.9 (3.4-17.3)	12.8 (7.3-21.5)	10.6 (5.7-18.9)	20.6 (14.0-29.2)	6.75	0.14	0.080	0.189

Supplementary Table 17 | Pairwise contrasts for the five omnibus-significant options (Fisher's exact tests, BH-corrected within the family of 30 pairwise tests; risk differences with Newcombe 95% CI).

Option	Contrast (A vs B)	% in A	% in B	Risk difference A – B, pp (Newcombe 95% CI)	Fisher p	BH-adjusted p	Significant
Advanced / specialised	no intention vs interested	17.5	23.3	-5.8 (-18.2 to 7.7)	0.421	0.505	no
Advanced / specialised	no intention vs non-routine	17.5	36.5	-19.0 (-32.0 to -4.4)	0.016	0.038	yes
Advanced / specialised	no intention vs routine	17.5	60.7	-43.3 (-54.7 to -28.6)	< 0.001	< 0.001	yes
Advanced / specialised	interested vs non-routine	23.3	36.5	-13.2 (-26.3 to 0.5)	0.067	0.112	no
Advanced / specialised	interested vs routine	23.3	60.7	-37.5 (-49.1 to -23.8)	< 0.001	< 0.001	yes
Advanced / specialised	non-routine vs routine	36.5	60.7	-24.3 (-37.1 to -10.1)	0.001	0.007	yes
Strategic	no intention vs interested	17.5	26.7	-9.3 (-21.9 to 4.6)	0.236	0.354	no
Strategic	no intention vs non-routine	17.5	29.4	-12.0 (-24.7 to 2.2)	0.121	0.192	no
Strategic	no intention vs routine	17.5	45.8	-28.3 (-40.3 to -13.9)	< 0.001	0.002	yes
Strategic	interested vs non-routine	26.7	29.4	-2.7 (-15.9 to 10.7)	0.736	0.761	no

Option	Contrast (A vs B)	% in A	% in B	Risk difference A – B, pp (Newcombe 95% CI)	Fisher p	BH-adjusted p	Significant
Strategic	interested vs routine	26.7	45.8	-19.1 (-31.5 to -5.4)	0.007	0.022	yes
Strategic	non-routine vs routine	29.4	45.8	-16.4 (-29.2 to -2.5)	0.025	0.050	no
Large language models	no intention vs interested	17.5	36.0	-18.6 (-31.5 to -4.0)	0.016	0.038	yes
Large language models	no intention vs non-routine	17.5	43.5	-26.1 (-39.0 to -11.1)	< 0.001	0.005	yes
Large language models	no intention vs routine	17.5	40.2	-22.7 (-34.8 to -8.5)	0.002	0.009	yes
Large language models	interested vs non-routine	36.0	43.5	-7.5 (-21.6 to 7.1)	0.351	0.457	no
Large language models	interested vs routine	36.0	40.2	-4.1 (-17.5 to 9.6)	0.655	0.708	no
Large language models	non-routine vs routine	43.5	40.2	+3.3 (-10.5 to 17.1)	0.661	0.708	no
Ethics and law	no intention vs interested	30.2	10.5	+19.7 (6.8 to 32.8)	0.003	0.011	yes
Ethics and law	no intention vs non-routine	30.2	12.9	+17.2 (4.0 to 30.6)	0.013	0.035	yes
Ethics and law	no intention vs routine	30.2	15.0	+15.2 (2.5 to 28.6)	0.030	0.056	no
Ethics and law	interested vs non-routine	10.5	12.9	-2.5 (-12.5 to 7.5)	0.642	0.708	no
Ethics and law	interested vs routine	10.5	15.0	-4.5 (-13.8 to 5.4)	0.395	0.494	no
Ethics and law	non-routine vs routine	12.9	15.0	-2.0 (-11.7 to 8.4)	0.835	0.835	no
Commercial AI vendors	no intention vs interested	4.8	22.1	-17.3 (-27.7 to -6.1)	0.004	0.014	yes
Commercial AI vendors	no intention vs non-routine	4.8	15.3	-10.5 (-20.2 to -0.2)	0.059	0.105	no
Commercial AI vendors	no intention vs routine	4.8	29.9	-25.1 (-34.9 to -13.7)	< 0.001	< 0.001	yes
Commercial AI vendors	interested vs non-routine	22.1	15.3	+6.8 (-5.0 to 18.4)	0.327	0.446	no
Commercial AI vendors	interested vs routine	22.1	29.9	-7.8 (-19.7 to 4.8)	0.252	0.360	no
Commercial AI vendors	non-routine vs routine	15.3	29.9	-14.6 (-25.7 to -2.6)	0.025	0.050	no

Supplementary Table 18 | Delivery-related options: observed differences and equivalence bounds.

Delivery option	Contrast (A vs B)	Risk difference A – B, pp (95% CI)	Largest absolute difference not excluded at the 90% level, pp
Self-paced online courses	no intention vs interested	-9.8 (-24.6 to 5.1)	22.3
Self-paced online courses	no intention vs non-routine	-10.6 (-25.4 to 4.2)	23.1
Self-paced online courses	no intention vs routine	-17.8 (-31.6 to -4.1)	29.5
Self-paced online courses	interested vs non-routine	-0.9 (-13.9 to 12.2)	11.8
Self-paced online courses	interested vs routine	-8.1 (-20.0 to 3.7)	18.1
Self-paced online courses	non-routine vs routine	-7.2 (-19.2 to 4.5)	17.2
Live webinars	no intention vs interested	-18.1 (-32.9 to -1.9)	30.7
Live webinars	no intention vs non-routine	-14.1 (-29.1 to 2.1)	26.8
Live webinars	no intention vs routine	-24.2 (-38.1 to -8.7)	36.1
Live webinars	interested vs non-routine	+4.1 (-10.7 to 18.6)	16.3
Live webinars	interested vs routine	-6.1 (-19.8 to 7.8)	17.6
Live webinars	non-routine vs routine	-10.2 (-23.8 to 3.9)	21.7
Nuggets (15-30 min)	no intention vs interested	-7.7 (-22.8 to 8.2)	20.5
Nuggets (15-30 min)	no intention vs non-routine	+3.6 (-11.5 to 18.9)	16.5
Nuggets (15-30 min)	no intention vs routine	-1.8 (-16.2 to 13.3)	14.0
Nuggets (15-30 min)	interested vs non-routine	+11.2 (-3.3 to 25.1)	23.0
Nuggets (15-30 min)	interested vs routine	+5.9 (-7.9 to 19.5)	17.4
Nuggets (15-30 min)	non-routine vs routine	-5.4 (-18.5 to 8.3)	16.5
Blended learning	no intention vs interested	-11.3 (-25.9 to 4.5)	23.7
Blended learning	no intention vs non-routine	+2.3 (-12.2 to 17.3)	14.9
Blended learning	no intention vs routine	-6.6 (-20.5 to 8.4)	18.4
Blended learning	interested vs non-routine	+13.6 (-0.8 to 27.2)	25.1
Blended learning	interested vs routine	+4.7 (-9.0 to 18.3)	16.2
Blended learning	non-routine vs routine	-8.9 (-21.7 to 4.6)	19.8
On-site sessions	no intention vs interested	-9.0 (-23.6 to 6.7)	21.4
On-site sessions	no intention vs non-routine	-0.0 (-14.6 to 15.1)	12.7
On-site sessions	no intention vs routine	-2.8 (-16.7 to 12.0)	14.6
On-site sessions	interested vs non-routine	+8.9 (-5.4 to 22.8)	20.6
On-site sessions	interested vs routine	+6.1 (-7.4 to 19.6)	17.5
On-site sessions	non-routine vs routine	-2.8 (-15.8 to 10.6)	13.8
Short formats (1-2 h)	no intention vs interested	-6.1 (-20.1 to 7.5)	17.9
Short formats (1-2 h)	no intention vs non-routine	-7.0 (-21.0 to 6.5)	18.7
Short formats (1-2 h)	no intention vs routine	-13.0 (-26.1 to -0.7)	24.0
Short formats (1-2 h)	interested vs non-routine	-0.9 (-13.0 to 11.2)	11.1
Short formats (1-2 h)	interested vs routine	-6.9 (-18.0 to 3.8)	16.2
Short formats (1-2 h)	non-routine vs routine	-6.0 (-17.1 to 4.6)	15.2
Half-day	no intention vs interested	-5.7 (-19.0 to 8.7)	17.0

Delivery option	Contrast (A vs B)	Risk difference A – B, pp (95% CI)	Largest absolute difference not excluded at the 90% level, pp
Half-day	no intention vs non-routine	-3.7 (-16.9 to 10.6)	14.9
Half-day	no intention vs routine	+0.7 (-11.4 to 14.2)	12.0
Half-day	interested vs non-routine	+2.0 (-11.2 to 15.1)	13.1
Half-day	interested vs routine	+6.4 (-5.7 to 18.7)	16.7
Half-day	non-routine vs routine	+4.4 (-7.5 to 16.6)	14.7
Longitudinal	no intention vs interested	-14.5 (-26.1 to -1.6)	24.2
Longitudinal	no intention vs non-routine	-5.4 (-16.2 to 6.6)	14.5
Longitudinal	no intention vs routine	-9.4 (-19.7 to 2.6)	18.1
Longitudinal	interested vs non-routine	+9.1 (-3.2 to 21.1)	19.2
Longitudinal	interested vs routine	+5.0 (-6.8 to 17.1)	15.2
Longitudinal	non-routine vs routine	-4.1 (-14.8 to 7.3)	13.1
Full-day	no intention vs interested	-3.7 (-13.3 to 7.0)	11.7
Full-day	no intention vs non-routine	-5.0 (-14.9 to 5.9)	13.2
Full-day	no intention vs routine	-3.3 (-11.9 to 7.2)	10.5
Full-day	interested vs non-routine	-1.3 (-11.5 to 8.8)	9.8
Full-day	interested vs routine	+0.4 (-8.6 to 10.1)	8.4
Full-day	non-routine vs routine	+1.7 (-7.5 to 11.7)	10.0
Application-oriented	no intention vs interested	-4.1 (-14.9 to 5.2)	13.0
Application-oriented	no intention vs non-routine	-7.6 (-17.9 to 0.9)	16.0
Application-oriented	no intention vs routine	+2.0 (-9.3 to 11.5)	10.0
Application-oriented	interested vs non-routine	-3.4 (-11.2 to 3.9)	9.8
Application-oriented	interested vs routine	+6.1 (-2.9 to 14.7)	13.2
Application-oriented	non-routine vs routine	+9.6 (1.4 to 17.6)	16.2

Script: analysis/adoption_groups.py (outputs A_omnibus.csv, B_pairwise.csv, C_risk_differences.csv, figS6_1.png).

Supplementary Note 7: Questionnaire

Questionnaire first in German (original) then in English (translated).

Items 8 to 11, 15, 20, 23 and 24 were displayed conditionally (branching on items 7, 14, 19 and 22, as noted at each item); all other items, including the barriers item (16), were displayed to all respondents.

Spelling errors present in the German original have been silently corrected in translation. The English wording of items reported in the manuscript follows the wording used there. Editorial notes on conditional display are given in italics in brackets in both language versions.

Fragebogen (Original, Deutsch)

Studie zu Schulungsangeboten zu KI im Gesundheitswesen

Vielen Dank, dass Sie an der Studie teilnehmen. (Microsoft Forms; * = Pflichtfeld. Originalschreibweise beibehalten, inkl. Tippfehler.)

Teil 1: Zustimmung zum Datenschutz und Einwilligung zur Studie

Vielen Dank, dass Sie an der Studie „Sensibilisierungs- und Schulungsangebote bezüglich Digitalisierung und Künstlicher Intelligenz (KI) innerhalb der deutschen Gesundheitsversorgung" teilnehmen.

Die Studie besteht aus drei Abschnitten: 1. Zustimmung zum Datenschutz und Einwilligung zur Studie; 2. Sozio-demographische Angaben; 3. Angaben zu bisherigen Digitalisierungs- und KI-Erfahrungen sowie Schulungen. Insgesamt werden für die Beantwortung der Fragen ca. 8–10 min benötigt. Bei weiteren Fragen oder Unklarheiten können Sie sich jederzeit an den Studienleiter Prof. Dr. Markus Schinle (markus.schinle@hs-offenburg.de) wenden.

1. Datenschutzerklärung zur Studie * (Datenschutzerklärung verlinkt.)

Einwilligungstext: „Ich habe die Datenschutzerklärung … gelesen und willige die damit verbundene Datenverarbeitung ein. Mir ist bewusst, dass die Einwilligung freiwillig ist und ohne Nachteile verweigert oder jederzeit auch ohne Angaben von Gründen widerrufen werden kann. …"

- ☐ Ich habe die Datenschutzerklärung gelesen und stimme der Datenschutzerklärung zu.

2. Teilnehmendeninformationen zur Studie * (Teilnehmendeninformation verlinkt.)
Einwilligungstext wie oben (Teilnahme + Datenverarbeitung).

- ☐ Ich habe die Teilnehmendeninformation gelesen und willige ein an der Studie teilzunehmen.

Teil 2: Sozio-demographische Fragen

3. Wie alt sind Sie? *

20–29 Jahre · 30–39 Jahre · 40–49 Jahre · 50–59 Jahre · 60–69 Jahre · 70 Jahre oder älter · keine Angabe

4. Welches Bundesland sind Sie tätig? *

Baden-Württemberg · Bayern · Brandenburg · Berlin · Bremen · Hamburg · Hessen · Mecklenburg-Vorpommern · Niedersachsen · Nordrhein-Westfalen · Rheinland-Pfalz · Saarland · Sachsen · Sachen-Anhalt · Schleswig-Holstein · Thüringen · keine Angabe · Sonstiges

5. Welchem Geschlecht ordnen Sie sich zu? *

männlich · weiblich · divers · keine Angabe

6. In welchem Umfeld arbeiten Sie hauptsächlich? *

Krankenhaus (stationär) · Krankenhaus (ambulant) · Praxis (Einzelpraxis) · Praxis (Gemeinschaftspraxis/Berufsausübungsgemeinschaft) · MVZ (Medizinisches Versorgungszentrum) · Pflegeheim/Ambulante Pflege · Rettungsdienst · Öffentlicher Gesundheitsdienst · Verwaltung/Management · Selbstverwaltung/Politik/Berufsverband · keine Angabe

7. Welcher Berufsgruppe ordnen sie sich zu? *

Medizinische Fachangestellte (MFA) · Professionelle Pflege · Ärzt:in · andere Healthcare Professionals · Sonstiges

8. In welcher Fachrichtung arbeiten Sie? *
(nur für Ärzt:innen; Verzweigung nach Frage 7)

Allgemeinmedizin · Allgemeinchirurgie · Anästhestiologie · Arbeitsmedizin · Augenheilkunde · Dermatologie · Gastrologie · Gynäkologie · Geriatrie · Hals-Nasen-Ohrenheilkunde · Hämatologie und Onkologie · Humangenetik · Hygiene- und Umweltmedizin · Innere Medizin · Intensivmedizin · Kardiologie · Kinder- und Jugendmedizin · Kinder- und Jugendpsychiatrie und -psychotherapie · Labormedizin · Mund- und Kieferchirugie · Neurochirugie · Neurologie · Nuklearmedizin · Öffentliches Gesundheitswesen · Orthopädie und Unfallchirurgie (inkl. plastische und rekonstruktive Chirurgie) · Pathologie · Pharamakologie · Psychiatrie · Psychosomatik · Radiologie · Rechtsmedizin · Strahlenmedizin · Thorax-Herz-Gefäßchirurgie · Transfusionsmedizin · Urologie · keine Angaben · Sonstiges

9. Wie viel Berufserfahrung haben Sie? (Gerechnet ab dem dritten Staatsexamen) *
(nur für Ärzt:innen; Verzweigung nach Frage 7)

0–4 Jahre · 5–9 Jahre · 10–14 Jahre · 15–19 Jahre · 20–25 Jahre · >25 Jahre · kein Angabe

10. Was ist ihre Berufsbezeichnung/Position? (Pflege und MFA) *
(nur nicht-ärztliche Berufsgruppen; Verzweigung nach Frage 7)

Altenpfleger:in · Auszubildende:r zur/zum Medizinischen Fachangestellten (MFA) · Auszubildende:r in der Pflege · Medizinische/r Fachangestellte:r (MFA)/Arzthelfer:in · Leitende/r MFA · MFA mit Zusatzqualifikation (z.B. VERAH, NäPa, Praxismanager:in, digi Manager:in) · Notfallsanitäter:in bzw. Mitarbeiter:in im Rettungsdienst · Pflegeassistent:in bzw. Pflegehelfer:in · Gesundheits- und Krankenpfleger:in, Pflegefachfrau/mann · Praxisanleiter:in in der Pflege · Pflegefachkraft mit Weiterbildung (z.B. Intensivpflege, Anästhesie, OP, Kinderintensiv) · Physician Assistant · Stationsleitung bzw. Teamleitung · Pflegedienstleitung · Sonstiges

11. Wie viel Berufserfahrung haben Sie? (Gerechnet ab dem Abschluss der Ausbildung/Studium) *
(nur nicht-ärztliche Berufsgruppen; Verzweigung nach Frage)

0–4 Jahre · 5–9 Jahre · 10–14 Jahre · 15–19 Jahre · 20–25 Jahre · >25 Jahre · kein Angabe

Teil 3: Fragen zu Digitalisierungs- und KI-Erfahrungen sowie diesbezüglicher Schulungen

KI-Definition (Vorbemerkung): „Eine einheitliche Definition für Künstliche Intelligenz gibt es nicht. Daher möchten wir uns für diesen Fragebogen auf die Definition des Deutschen Ärzteblatts stützen: ‚dass es sich um ein Teilgebiet der Informatik sowie die Fähigkeit von Systemen handelt, selbstständig kognitive Funktionen wie Lernen, Denken, Planen und Problemlösung zu erfüllen, für die normalerweise menschliche Intelligenz nötig ist. Dabei unterscheidet sie sich (noch) in zentralen Eigenschaften wie dem Bewusstsein von menschlicher Intelligenz.'" (Deutsches Ärzteblatt, Sonderausgabe KI 2025)

12. Wie schätzen Sie Ihre digitale Kompetenz im Alltag (nicht beruflicher Kontext) ein? (1 = sehr niedrig … 5 = sehr hoch)

1 · 2 · 3 · 4 · 5

13. Wie häufig nutzen Sie folgende digitale Werkzeuge in Ihrem Berufsalltag? *(Matrix)*
Skala: täglich · wöchentlich · monatlich · seltener · möchte ich, aber nicht verfügbar · unklar, was das ist

Items (19): Praxisverwaltungssysteme (PVS)/Krankenhausinformationssystem (KIS) · Elektronische Patientenakte (ePA) · Elektronische Medikationsliste (eML/eMP) · Elektronische Arbeitsunfähigkeit (eAU) · Bildbetrachtungssysteme (z.B. PACS) · Laborinformationssysteme (LIS) · Differentialdiagnostik bzw. Clinical-Decision-Support (CDSS) · Digitale Gesundheitsanwendungen (DiGA) · Gesundheits-Apps für die Patientenversorgung · Telefon-Assistenzsysteme (z.B. für Terminvergabe) · Videosprechstunde · Tele-Visite (z.B. Intensivstation/Stroke-Unit/Pflegeheim) · Tele-/Home-Monitoring · Spracherkennung zur automatisierten Behandlungs-Dokumentation · Digitale Patientenvoranmeldung (NIDA, IVENA, Pulsation) in Notaufnahme/im Rettungsdienst · Ersteinschätzungs-Tools (wie z.B. von docyet, ada, SMeD) · Arzneimittel Therapiesicherheitssystem (AMTS) – Hinweis bei Medikamenten Anordnung · Automatisierte Dienstplanung · Wissensmanagement (z.B. Amboss, Elsevier, Thieme)

14. Stellt Ihnen Ihr Arbeitgeber KI-gestütze Anwendungen zur Verfügung? *

Ja · Nein · Weiß ich nicht / Kann ich nicht einschätzen

15. In welcher Form werden diese KI-Tools bereitgestellt? *
(nur bei „Ja" in Frage 14)

Als eigenständige Tools oder Programme (z.B. KI-Modelle oder Chatbots, Spracherkennung, usw.) · In bestehender medizinischer Software integriert (z.B. Diagnosesysteme, Entscheidungsunterstützung) · In Erprobung/Pilotphase (von externer Software) · Inhouse entwickelte KI-Modelle · Keine KI-Tools verfügbar · Weiß nicht · Sonstiges

16. Was sind die Hauptgründe, warum Sie derzeit keine KI-basierten Anwendungen nutzen? * — *Wählen Sie höchstens 3 Optionen aus.*

Nicht verfügbar · Sicherheits-/Datenschutzbedenken · Fehlendes Wissen/Schulungen · Keine wahrgenommene Nützlichkeit/Zeitgewinn · Mangelende Integrationsmöglichkeiten in bestehenden Systemen · fehlende Kapazitäten/Zeitmangel für Implementierung · Sonstiges

17. Wie schätzen Sie Ihre digitale Kompetenz im beruflichen Kontext ein? (1 = sehr niedrig … 5 = sehr hoch) *

1 · 2 · 3 · 4 · 5

18. Wie bewerten Sie die Unterstützung (z.B. Schulungen, Anleitungen) bei der Einführung neuer digitaler Werkzeuge an Ihrem Arbeitsplatz? * —

Freitext

19. Nutzen Sie bereits KI-basierte Anwendungen in Ihrem beruflichen Alltag *

Ja, regelmäßig (mehrfach pro Woche) · Ja, gelegentlich (einmal pro Woche) · Ja, selten/experimentell (einmal pro Monat) · Nein, aber geplant/Interesse vorhanden · Nein, und aktuell nicht geplant · Ich weiß es nicht/Bin mir unsicher · Keine Antwort

20. Für welche der folgenden Bereiche nutzen Sie KI-Anwendungen? (Mehrfachnennungen möglich) *
(nur bei „Ja, …" oder „Ich weiß es nicht/Bin mir unsicher" in Frage 19)

Diagnostik und Befund (z.B. Bildanalyse, Mustererkennung, EKG-Auswertung etc.) · Behandlungsplanung und Therapieentscheidungen (z.B. Mediaktionsmanagement, personalisierte Therapieempfehlungen) · Administrative Aufgaben (z.B. Dokumentation wie Arztbrief schreiben, Abrechnung, etc.) · Forschung (z.B. Datenanalyse, Literaturrecherche, Simulationen) · Patientenkommunikation und Service (z.B. Chatbots, Terminvereinbarung, automatisierte Anamnese-Tools) · keine Antwort · Sonstiges

21. Wie setzen Sie sich im privaten Umfeld mit dem Thema KI auseinander? (Mehrfachnennung möglich) *

Ich informiere mich aktiv (z.B. Fachartikel, Kurse, Podcasts) · Ich lese gelegentlich Nachrichten/Artikel dazu · Ich nutze KI-Anwendungen privat (z.B. ChatGPT, KI-Bildgeneratoren) · Über soziale Medien oder Netzwerke (z.B. LinkedIn) · Eher gar nicht/nur passiv · Keine Antwort · Sonstiges

22. Hatten Sie im beruflichen Kontext bereits Schulungen spezifisch zum Thema KI erhalten? * Ja · Nein · Keine Antwort

23. Welche Art von KI-Schulung war das? (Mehrfachnennung möglich) *
(Nur bei „Ja" in Frage 22)
Kurzvortrag/Informationsveranstaltung (<2 Stunden) · Workshop (wenige Stunden bis 1 Tag) · Mehrtägiger Kurs/Zertifikatslehrgang · Online-Modul/E-learning · Integrierte in anderen Fortbildungen · Keine Antwort · Sonstiges

24. Wie bewerten Sie die Nützlichkeit dieser Schulung(en) für Ihre Arbeit? (1 = gar nicht brauchbar … 5 = sehr nützlich) *
(Nur bei „Ja" in Frage 22)

1 · 2 · 3 · 4 · 5

25. Welche Formate für KI-Schulungen würden Sie bevorzugen? (Mehrfachnennungen möglich) *

Online-Kurse (flexibel, Selbstlernformat) · Live Online-Webinare (interaktiv) · Präsenzveranstaltungen/Workshop vor Ort · Blended Learning (Kombination aus Online und Präsenz) · Kurze Informations-Nuggets (z.B. 15–30 min Videos) · Kein Präferenz · Sonstiges

26. Welchen Umfang/Dauer für KI-Schulungen fänden Sie ideal (Mehrfachnennung möglich)? *

Kurzformate (z.B. 1–2 Stunden) · Halbtägige Veranstaltungen · Ganztägige Veranstaltungen · Mehrtägige Kurse · Längerfristige Weiterbildungen (z.B. über mehrere Wochen/Monate verteilt) · Keine Präferenz · Sonstiges

27. Auf welchem Niveau sollten die KI-Schulungen für Sie idealerweise ansetzen? (Mehrfachnennung möglich) *

Einführend/Grundlagen vermittelnd (Fokus auf Verständnis der Basiskonzepte) · Anwendungsorientiert/Praxisbezogen (Fokus auf Nutzung spezifischer Tools und Methoden im Alltag) · Vertiefend/Spezialisiert (Fokus auf detaillierte Kenntnisse spezifischer KI-Methoden oder komplexer Anwendungsfälle) · Strategisch/Managementbezogen (Fokus auf Implementierung) · Keine Antwort · Sonstiges

28. Welche der folgenden Themenbereiche für KI-Schulungen wären für Sie besonders relevant? * — *Wählen Sie höchstens 3 Optionen aus.*

KI in der diagnostischen Bildgebung · KI-gestützte Diagnosesysteme · KI in der Therapieplanung und personalisierter Medizin · Umgang mit großen Sprachmodellen (z.B. ChatGPT) im medizinischen Kontext · Datenmanagement und Datenschutz bei KI-Anwendungen · Ethische und rechtliche Rahmenbedingungen für KI in der Medizin · Kritische Berwertung von KI-Anwendungen und Studien · Praktische Implementierung von KI-Tools im Arbeitsalltag · KI in der medizinischen Forschung · KI und Patienteninteraktion und -kommunikation · Keine Antwort · Sonstiges

29. Welche Organisation oder Institution sollte dieses Angebot anbieten (Mehrfachnennung möglich)? *

Ärztekammern · Arbeitgeber (Krankenhäuser/Arztpraxen) · Berufsverbände (z.B. Hartmannbund e.V., Marburger Bund) · Fachgesellschaften (z.B. DGIM, DEGAM, DGOU) · Industrieanbieter von KI-Software · Kassenärztliche Vereinigungen · Unabhängige private Weiterbildungsinstitute · Universitäten/Hochschulen · Keine Präferenz · Sonstiges

30. Welche Gründe halten Sie davon ab, sich zu KI in der Medizin weiterzubilden? * — *Freitext*

31. Haben Sie weitere Anmerkungen oder Wünsche bezüglich KI-Schulungen für Ihre Berufsgruppe? * — *Freitext*

32. Kennen Sie Fälle von Schatten-KI in ihrem beruflichen Umfeld? * *Definition:* „Unter Schatten-KI versteht man die Verwendung von KI-basierten Anwendungen oder Tools durch Ärzt:innen, die ohne offizielle Genehmigung oder ohne Aufsicht durch die zuständigen IT- oder Datenschutzverantwortlichen erfolgt. Beispiel: Die Nutzung von ChatGPT auf dem privaten Endgerät zur Diagnose-Stellung."

Ja, habe selbst schon mal KI ohne formelle Genehmigung genutzt. · Ja, habe andere schon mal KI ohne formelle Genehmigung nutzen sehen. · Nein, mir ist kein Fall bekannt · keine Antwort

33. Haben Sie von ihrem Arbeitgeber eine Leitlinie oder ähnliches zum Einsatz und der Nutzung von Künstlicher Intelligenz, insbesondere Large Language Models (ChatGPT, Microsoft Co-Pilot, Claude, etc.) erhalten? *

Ja · Nein · Weiß ich nicht · Sonstiges

Questionnaire in English (translated)

This is an English translation of the original German questionnaire administered via Microsoft Forms. Item numbering follows the original. Question type and any selection limits are reproduced as displayed.

Part 1: Consent to data protection and participation in the study

Thank you for taking part in the study 'Awareness and training offerings regarding digitalisation and artificial intelligence (AI) within German healthcare provision'.

The study consists of three sections: (1) consent to data protection and participation in the study; (2) socio-demographic information; (3) information on previous digitalisation and AI experiences and related training.

Answering the questions takes approximately 8 to 10 minutes in total.

If you have further questions or uncertainties, you may contact the study leader [name and institutional email redacted] at any time.

Q1. I have read the data protection declaration and agree to it.

Single choice. Required. A link to the data protection declaration was provided in the original [link redacted].

Consent statement shown to respondents: I have read the data protection declaration for participation in the study and consent to the associated data processing. I am aware that consent is voluntary and may be refused without disadvantage or withdrawn at any time without giving reasons. I understand that, in the event of withdrawal, the lawfulness of the processing carried out on the basis of consent up to that point is not affected, and that to withdraw I can simply contact the person named in the participant information and the data protection declaration.

Q2. I have read the participant information and consent to take part in the study.

Single choice. Required. A link to the participant information was provided in the original [link redacted].

Consent statement shown to respondents: I have read the participant information and the data protection declaration for participation in the study and consent to participation and the associated data processing. The voluntariness and withdrawal conditions stated for item 1 apply equally.

Part 2: Socio-demographic questions

Q3. How old are you?

Single choice. Required.

- 20 to 29 years
- 30 to 39 years
- 40 to 49 years
- 50 to 59 years
- 60 to 69 years
- 70 years or older
- No answer

Q4. In which federal state do you work?

Single choice. Required.

- Baden-Wuerttemberg
- Bavaria
- Brandenburg

- Berlin
- Bremen
- Hamburg
- Hesse
- Mecklenburg-Western Pomerania
- Lower Saxony
- North Rhine-Westphalia
- Rhineland-Palatinate
- Saarland
- Saxony
- Saxony-Anhalt
- Schleswig-Holstein
- Thuringia
- No answer
- Other

Q5. Which gender do you identify with?

Single choice. Required.

- Male
- Female
- Diverse
- No answer

Q6. In which setting do you mainly work?

Single choice. Required.

- Hospital (inpatient)
- Hospital (outpatient)
- Practice (single practice)
- Practice (group practice / joint medical practice)
- Medical care centre (MVZ)
- Nursing home / outpatient care
- Emergency medical service
- Public health service
- Administration / management
- Self-governance / politics / professional association
- No answer

Q7. Which professional group do you belong to?

Single choice. Required.

- Medical assistant (MFA)
- Professional nursing

- Physician
- Other healthcare professionals
- Other

Q8. In which specialty do you work?

Single choice. Required. Conditional on item 7: displayed only to respondents selecting 'Physician'.

- General medicine
- General surgery
- Anaesthesiology
- Occupational medicine
- Ophthalmology
- Dermatology
- Gastroenterology
- Gynaecology
- Geriatrics
- Otorhinolaryngology (ENT)
- Haematology and oncology
- Human genetics
- Hygiene and environmental medicine
- Internal medicine
- Intensive care medicine
- Cardiology
- Paediatric and adolescent medicine
- Child and adolescent psychiatry and psychotherapy
- Laboratory medicine
- Oral and maxillofacial surgery
- Neurosurgery
- Neurology
- Nuclear medicine
- Public health
- Orthopaedics and trauma surgery (incl. plastic and reconstructive surgery)
- Pathology
- Pharmacology
- Psychiatry
- Psychosomatic medicine
- Radiology
- Forensic medicine
- Radiation medicine
- Thoracic, cardiac and vascular surgery

- Transfusion medicine
- Urology
- No answer
- Other

Q9. How much professional experience do you have? (Counted from the third state examination)

Single choice. Required. Conditional on item 7: displayed only to respondents selecting 'Physician'.

- 0 to 4 years
- 5 to 9 years
- 10 to 14 years
- 15 to 19 years
- 20 to 25 years
- More than 25 years
- No answer

Q10. What is your job title / position? (Nursing and medical assistants)

Single choice. Required. Conditional on item 7: displayed only to respondents not selecting 'Physician'. Answered by one respondent who was recoded as a physician during data cleaning (Supplementary Note 1).

- Geriatric nurse
- Trainee medical assistant (MFA)
- Nursing trainee
- Medical assistant (MFA) / doctor's assistant
- Lead medical assistant
- Medical assistant with additional qualification (e.g. VERAH, NaePa, practice manager, digital manager)
- Paramedic or emergency medical service staff
- Nursing assistant or nursing aide
- Registered nurse (health and nursing care professional)
- Practice instructor in nursing
- Specialist nurse with further training (e.g. intensive care, anaesthesia, operating theatre, paediatric intensive care)
- Physician assistant
- Ward manager or team leader
- Nursing service manager
- Other

Q11. How much professional experience do you have? (Counted from completion of training / studies)

Single choice. Required. Conditional on item 7: displayed only to respondents not selecting 'Physician'. Answered by one respondent who was recoded as a physician during data cleaning (Supplementary Note 1).

- 0 to 4 years
- 5 to 9 years

- 10 to 14 years
- 15 to 19 years
- 20 to 25 years
- More than 25 years
- No answer

Part 3: Questions on digitalisation and AI experiences and related training

Definition shown to respondents: There is no uniform definition of artificial intelligence. For this questionnaire we therefore rely on the definition of the Deutsches Aerzteblatt, namely that AI is a subfield of computer science and the ability of systems to perform independently cognitive functions such as learning, thinking, planning and problem-solving that normally require human intelligence, while (still) differing from human intelligence in central characteristics such as consciousness. Source: Deutsches Aerzteblatt, AI special issue 2025.

Q12. How would you rate your digital competence in everyday life (non-professional context)?

Rating scale 1 to 5, where 1 = very low and 5 = very high.

Q13. How frequently do you use the following digital tools in your working day?

Matrix question. One response selected per row from six options: daily; weekly; monthly; rarely; 'I would like to, but it is not available'; 'unclear what this is'.

Items (19): practice management systems (PVS) / hospital information system (HIS); electronic patient record (ePA); electronic medication list (eML/eMP); electronic certificate of incapacity for work (eAU); image viewing systems (e.g. PACS); laboratory information systems (LIS); differential diagnostics / clinical decision support (CDSS); digital health applications (DiGA); health apps for patient care; telephone assistance systems (e.g. for appointment scheduling); video consultation; tele-visit (e.g. intensive care unit / stroke unit / nursing home); tele- / home monitoring; speech recognition for automated treatment documentation; digital patient pre-registration (NIDA, IVENA, Pulsation) in the emergency department / emergency medical service; initial assessment / triage tools (e.g. docyet, ada, SMeD); drug therapy safety system (AMTS): alerts when prescribing medication; automated duty / shift scheduling; knowledge management (e.g. Amboss, Elsevier, Thieme).

Q14. Does your employer provide you with AI-based applications?

Single choice. Required.

- Yes
- No
- I don't know / cannot say

Q15. In what form are these AI tools provided?

Single choice (one form per respondent in the export). Required. Conditional on item 14: displayed only to respondents who answered 'yes'.

- As standalone tools or programs (e.g. AI models or chatbots, speech recognition, etc.)
- Integrated into existing medical software (e.g. diagnostic systems, decision support)
- In trial / pilot phase (external software)
- In-house developed AI models
- No AI tools available
- Don't know
- Other

Q16. What are the main reasons why you currently do not use AI-based applications?

Multiple choice, select at most 3. Required. Displayed to all respondents; analysed for definite non-users (n = 149).

- Not available
- Security / data protection concerns
- Lack of knowledge / training
- No perceived usefulness / time saving
- Lack of integration options in existing systems
- Lack of capacity / time for implementation
- Other

Q17. How would you rate your digital competence in a professional context?

Rating scale 1 to 5, where 1 = very low and 5 = very high. Required.

Q18. How do you rate the support (e.g. training, instructions) when introducing new digital tools at your workplace?

Free text. Required.

Q19. Do you already use AI-based applications in your professional daily routine?

Single choice. Required. Provides the AI-use frequency categories reported in the manuscript.

- Yes, regularly (several times per week)
- Yes, occasionally (once per week)
- Yes, rarely / experimentally (once per month)
- No, but planned / interested
- No, and not currently planned
- I don't know / am unsure
- No answer

Q20. For which of the following areas do you use AI applications? (Multiple answers possible)

Multiple choice. Required. Conditional on item 19: displayed to respondents reporting AI use and to those answering 'I don't know / am unsure'.

- Diagnostics and findings (e.g. image analysis, pattern recognition, ECG evaluation, etc.)
- Treatment planning and therapy decisions (e.g. medication management, personalised therapy recommendations)
- Administrative tasks (e.g. documentation such as writing physician letters, billing, etc.)
- Research (e.g. data analysis, literature search, simulations)
- Patient communication and service (e.g. chatbots, appointment scheduling, automated history-taking tools)
- No answer
- Other

Q21. How do you engage with the topic of AI in your private life? (Multiple answers possible)

Multiple choice. Required.

- I actively inform myself (e.g. specialist articles, courses, podcasts)

- I occasionally read news / articles about it
- I use AI applications privately (e.g. ChatGPT, AI image generators)
- Via social media or networks (e.g. LinkedIn)
- Hardly at all / only passively
- No answer
- Other

Q22. Have you already received training specifically on the topic of AI in a professional context?

Single choice. Required.

- Yes
- No
- No answer

Q23. What type of AI training was it? (Multiple answers possible)

Multiple choice. Required. Conditional on item 22: displayed only to respondents who answered 'yes'.

- Short presentation / information session (under 2 hours)
- Workshop (a few hours up to 1 day)
- Multi-day course / certificate programme
- Online module / e-learning
- Integrated into other continuing education
- No answer
- Other

Q24. How do you rate the usefulness of this training for your work?

Rating scale 1 to 5, where 1 = not useful at all and 5 = very useful. Required. Conditional on item 22: displayed only to respondents who answered 'yes'.

Q25. Which formats for AI training would you prefer? (Multiple answers possible)

Multiple choice. Required.

- Online courses (flexible, self-study format)
- Live online webinars (interactive)
- In-person events / on-site workshop
- Blended learning (combination of online and in-person)
- Short information nuggets (e.g. 15 to 30 minute videos)
- No preference
- Other

Q26. What scope / duration for AI training would you consider ideal? (Multiple answers possible)

Multiple choice. Required.

- Short formats (e.g. 1 to 2 hours)
- Half-day events
- Full-day events
- Multi-day courses

- Longer-term continuing education (e.g. spread over several weeks / months)
- No preference
- Other

Q27. At what level should the AI training ideally be pitched for you? (Multiple answers possible)

Multiple choice. Required.

- Introductory / foundational (focus on understanding basic concepts)
- Application-oriented / practical (focus on using specific tools and methods in daily work)
- Advanced / specialised (focus on detailed knowledge of specific AI methods or complex use cases)
- Strategic / management-oriented (focus on implementation)
- No answer
- Other

Q28. Which of the following topic areas for AI training would be particularly relevant for you?

Multiple choice, select at most 3. Required.

- AI in diagnostic imaging
- AI-supported diagnostic systems
- AI in therapy planning and personalised medicine
- Handling large language models (e.g. ChatGPT) in the medical context
- Data management and data protection in AI applications
- Ethical and legal frameworks for AI in medicine
- Critical appraisal of AI applications and studies
- Practical implementation of AI tools in daily work
- AI in medical research
- AI and patient interaction and communication
- No answer
- Other

Q29. Which organisation or institution should provide this offering? (Multiple answers possible)

Multiple choice. Required.

- Medical chambers (Aerztekammern)
- Employers (hospitals / medical practices)
- Professional associations (e.g. Hartmannbund e.V., Marburger Bund)
- Specialist societies (e.g. DGIM, DEGAM, DGOU)
- Industry providers of AI software
- Associations of Statutory Health Insurance Physicians (Kassenaerztliche Vereinigungen)
- Independent private continuing education institutes
- Universities / higher education institutions
- No preference
- Other

Q30. What reasons keep you from pursuing further training on AI in medicine?

Free text. Required.

Q31. Do you have any further comments or requests regarding AI training for your professional group?

Free text. Required.

Q32. Are you aware of cases of Shadow AI in your professional environment?

Single choice. Required. Analysed in a separate manuscript; in this article the first response option is used only as a marker of exposure (Supplementary Note 10).

Definition shown to respondents: Shadow AI refers to the use of AI-based applications or tools by physicians without official authorisation or without oversight by the responsible IT or data protection officers. Example: using ChatGPT on a private device to make a diagnosis.

- Yes, I have used AI myself without formal authorisation before.
- Yes, I have seen others use AI without formal authorisation before.
- No, I am not aware of any such case.
- No answer

Q33. Have you received a guideline or similar from your employer on the deployment and use of artificial intelligence, in particular large language models (ChatGPT, Microsoft Co-Pilot, Claude, etc.)?

Single choice. Not analysed in this article.

- Yes
- No
- I don't know
- Other

Supplementary Note 8: Latent class analysis of digital application use: item screen, model selection, convergence, classification certainty, the four-class solution, and sensitivity to coding and item set

Scope of LCA

The primary model (Fig. 2) treats each of 16 application items as having four states ('I would like to, but it is not available'; 'unclear what this is'; rarely to monthly; weekly to daily) and retains three classes. This note reports the applicability screen that excluded three of the 19 items (Supplementary Table 19), the diagnostics of the primary model (Supplementary Tables 20 to 24), its four-class solution (Supplementary Table 25), the stability of the three use profiles across codings and item sets, including the specification with 19 items, three levels ('not available' and 'unclear' combined) and 20 starts (Supplementary Table 26), the two partitions of the core-set profile that compete across specifications (Supplementary Table 27), sector-stratified results (Supplementary Table 28), and the person-level analysis of unfamiliarity (Supplementary Table 29). All models use StepMix 3.0 (categorical measurement model, expectation–maximisation, absolute tolerance 10^{-10}, maximum 1,000 iterations, random_state 42); classes are labelled by their response pattern (Supplementary Table 23), presented in the same order throughout, and assigned by maximum posterior probability. Entropy is the relative entropy $1 - \Sigma(-p \ln p)/(n \ln k)$; aBIC = $-2LL + p \cdot \ln((n + 2)/24)$; ARI, adjusted Rand index between modal partitions.

The four-state coding and the item screen were introduced in revision. The submitted analysis used 19 items with 'not available' and 'unclear what this is' combined, and under that specification the BIC favoured four classes. It is retained in Supplementary Table 26 as a sensitivity analysis.

Item Screen

For each item, Supplementary Table 19 gives the share of respondents using it weekly or daily overall and by sector, the share reporting any use, and the shares reporting it as not available or not recognised. Three items were excluded from the latent class analysis: triage tools (patient-facing; used weekly or daily by 1.6% of ambulatory and 2.7% of hospital respondents; not recognised by 39.9%), tele-rounds (wording tied to intensive care, stroke units and nursing homes; 5.8% and 4.0%) and digital pre-registration (wording tied to emergency departments and emergency medical services; 2.6% of ambulatory respondents, although 25.3% of hospital respondents). Clinical decision support was retained: it is used weekly or daily by few respondents (11.8%) but is applicable to all, and its 'unclear' rate (41.9%) is a finding rather than a property of the setting. The excluded items remain in the item-level descriptives.

Supplementary Table 19 | Applicability screen of the 19 application items (LCA sample, n = 356)

Item	Used weekly–daily, all %	ambulatory %	hospital %	Any use (rarely–daily) %	Not available %	Unclear what this is %	Setting named in item wording	Excluded
Triage tools	2.0	1.6	2.7	35.7	23.0	39.9		yes
Tele-rounds	4.8	5.8	4.0	52.2	36.0	11.0	yes	yes
Automated duty scheduling	7.6	7.9	6.0	49.2	33.7	15.7		
Tele-/home monitoring	9.3	12.6	4.7	52.5	35.7	11.2		
Clinical decision support	11.8	10.5	14.0	36.2	21.9	41.9		

Item	Used weekly–daily, all %	ambulatory %	hospital %	Any use (rarely–daily) %	Not available %	Unclear what this is %	Setting named in item wording	Excluded
Video consultation	12.1	17.8	5.3	58.1	38.2	2.8		
Digital pre-registration	13.2	2.6	25.3	50.0	24.4	23.6	yes	yes
Phone assistance	15.7	23.6	6.7	54.2	39.0	6.5		
DiGA	16.6	19.9	12.7	68.0	16.3	15.7		
Health apps	16.9	20.4	12.0	73.0	19.1	7.9		
Speech recognition	28.7	16.8	44.7	59.0	37.6	3.4		
Medication safety (AMTS)	36.2	35.1	40.7	61.8	19.1	18.5		
Electronic patient record (ePA)	45.2	62.3	26.7	65.2	34.3	0.6		
e-medication list	47.2	52.4	44.7	72.2	22.5	5.1		
e-sick note (eAU)	54.2	81.2	23.3	80.6	17.1	1.7		
PACS / imaging	55.3	38.7	78.7	79.8	11.8	8.4		
Knowledge management	61.2	49.2	78.0	88.5	6.2	4.8		
Laboratory IS	70.5	67.5	78.0	82.6	5.9	11.0		
Practice/hospital IS	89.6	93.7	88.7	94.9	2.2	2.8		

Model selection, convergence and classification certainty of the primary model

The BIC selects three classes; the AIC, aBIC and the bootstrap likelihood-ratio test do not terminate within the range examined (Supplementary Table 20). The three-class solution is reproduced exactly from 100 starts under three seeds, with six to seven distinct local maxima within five log-likelihood units of the optimum, one of which (0.65 units below the optimum) divides the routine-use and core-set profiles differently (see 'Unanswered cells and competing solutions'). The four-class solution, by contrast, is reached by one to three of 100 starts and has 21 to 22 competing local maxima (Supplementary Table 21). Classification certainty is high in both solutions (Supplementary Table 22).

The three profiles are described in Supplementary Tables 23 and 24: the wide-range, routine-use profile uses ten of the sixteen applications predominantly weekly or daily; the wide-range, occasional-use profile uses a similarly broad set predominantly rarely to monthly (for the peripheral applications the probability of rare-to-monthly use lies between 0.74 and 0.90); the core-set profile uses the practice or hospital information system, the laboratory and imaging systems and knowledge-management tools routinely and reports the remaining applications predominantly as not available, with clinical decision support, DiGA, medication-safety alerts and automated duty scheduling also frequently not recognised.

Supplementary Table 20 | Model selection, primary specification (16 items, four levels, 100 starts, seed 42)

k	Log-likelihood	Parameters	AIC	BIC	aBIC	Entropy	Smallest class n (%)	Class sizes	BLRT k vs k+1: 2ΔLL (95th pct of null), p
1	-6304.2	48	12704.4	12890.4	12738.1	–	–	356	–
2	-5874.6	97	11943.3	12319.1	12011.4	0.883	177 (49.7)	179/177	356.6 (107.1), p = 0.005
3	-5696.3	146	11684.7	12250.4	11787.2	0.902	69 (19.4)	165/122/69	279.2 (109.0), p = 0.005
4	-5556.7	195	11503.5	12259.1	11640.4	0.911	61 (17.1)	119/89/87/61	206.0 (101.4), p = 0.005
5	-5453.7	244	11395.4	12340.9	11566.8	0.916	52 (14.6)	115/75/59/55/52	–

Supplementary Table 21 | Convergence, primary specification

k	Specification	Starts reaching the maximum	Distinct local maxima within 5 LL units	Maximum log-likelihood	ARI with reference partition
3	100 starts, seed 42 (reference)	22 of 100	7	-5696.33	
3	100 starts, seed 1	20 of 100	6	-5696.33	1.000
3	100 starts, seed 2026	21 of 100	6	-5696.33	1.000

k	Specification	Starts reaching the maximum	Distinct local maxima within 5 LL units	Maximum log-likelihood	ARI with reference partition
4	100 starts, seed 42 (reference)	2 of 100	21	-5556.73	
4	100 starts, seed 1	3 of 100	22	-5556.73	1.000
4	100 starts, seed 2026	1 of 100	21	-5556.73	1.000

Note: ARI with the seed-42 solution of the same k.

Supplementary Table 22 | Classification certainty

k	Class	n	Mean posterior of assigned class	Minimum	Assigned with p ≥ 0.80 (%)	Assigned with p ≥ 0.90 (%)
3	Wide range, routine use	69	0.936	0.535	85.5	76.8
3	Wide range, occasional use	122	0.971	0.508	95.9	91.8
3	Core set, routine use	165	0.962	0.516	92.1	89.7
3	all	356	0.960	0.508	92.1	87.9
4	Wide range, routine use	61	0.932	0.539	88.5	77.0
4	Wide range, occasional use	119	0.970	0.511	95.8	90.8
4	Core set, routine use (ambulatory profile)	87	0.951	0.429	94.2	82.8
4	Core set, routine use (hospital profile)	89	0.949	0.368	92.1	86.5
4	all	356	0.954	0.368	93.3	85.4

Average posterior probability matrix (rows: assigned class; columns: mean posterior probability of each class), three-class solution:

	P(Wide range, routine use)	P(Wide range, occasional use)	P(Core set, routine use)
assigned: Wide range, routine use	0.936	0.030	0.034
assigned: Wide range, occasional use	0.013	0.971	0.016
assigned: Core set, routine use	0.020	0.018	0.962

Supplementary Table 23 | Profiles of the primary model (three classes)

Profile	n	%	AI use % (n defined)	Employer provision %	Prior AI training %	Hospital %	Ambulatory %	Item responses: weekly–daily %	rarely–monthly %	not available %	unclear %
Wide range, routine use	69	19.4	83.1 (65)	56.5	26.1	26.1	72.5	62.4	22.2	10.3	5.0
Wide range, occasional use	122	34.3	54.9 (113)	24.6	15.6	42.6	49.2	28.1	56.1	8.4	7.2
Core set, routine use	165	46.4	45.9 (157)	20.6	14.5	48.5	49.1	31.1	16.4	38.1	13.9

χ^2 across profiles: AI use $\chi^2(2) = 26.0$, $p < 0.001$ (n = 335); employer provision $\chi^2(2) = 32.2$, $p < 0.001$; prior training $\chi^2(2) = 4.9$, $p = 0.087$.

Supplementary Table 24 | Conditional response probabilities, primary model (16 items × three classes)

Item	Routine: P(weekly–daily)	Routine: P(not available)	Routine: P(unclear)	Occasional: P(weekly–daily)	Occasional: P(not available)	Occasional: P(unclear)	Core set: P(weekly–daily)	Core set: P(not available)	Core set: P(unclear)
Practice/hospital IS	0.97	0.01	0.00	0.88	0.00	0.01	0.87	0.04	0.06
Electronic patient record (ePA)	0.78	0.10	0.01	0.37	0.28	0.00	0.38	0.49	0.01
e-medication list	0.75	0.06	0.07	0.38	0.16	0.04	0.42	0.34	0.05
e-sick note (eAU)	0.86	0.07	0.03	0.44	0.05	0.00	0.48	0.32	0.02
PACS / imaging	0.65	0.07	0.04	0.53	0.02	0.10	0.53	0.21	0.09
Laboratory IS	0.96	0.01	0.00	0.59	0.04	0.14	0.68	0.10	0.13
Clinical decision support	0.38	0.10	0.23	0.05	0.12	0.38	0.06	0.34	0.53
DiGA	0.59	0.01	0.03	0.03	0.08	0.10	0.08	0.28	0.26
Health apps	0.56	0.04	0.03	0.04	0.07	0.02	0.10	0.34	0.14
Phone assistance	0.48	0.17	0.06	0.09	0.09	0.01	0.07	0.71	0.11
Video consultation	0.44	0.29	0.00	0.04	0.09	0.00	0.05	0.66	0.06
Tele-/home monitoring	0.29	0.26	0.07	0.03	0.03	0.04	0.06	0.65	0.18
Speech recognition	0.45	0.26	0.03	0.23	0.15	0.02	0.26	0.59	0.04
Medication safety (AMTS)	0.74	0.09	0.10	0.24	0.03	0.16	0.30	0.36	0.24
Automated duty scheduling	0.26	0.10	0.07	0.03	0.12	0.11	0.03	0.63	0.23

Item	Routine: P(weekly–daily)	Routine: P(not available)	Routine: P(unclear)	Occasional: P(weekly–daily)	Occasional: P(not available)	Occasional: P(unclear)	Core set: P(weekly–daily)	Core set: P(not available)	Core set: P(unclear)
Knowledge management	0.83	0.00	0.01	0.52	0.04	0.02	0.59	0.12	0.08

P(rarely–monthly) is the remainder to 1.

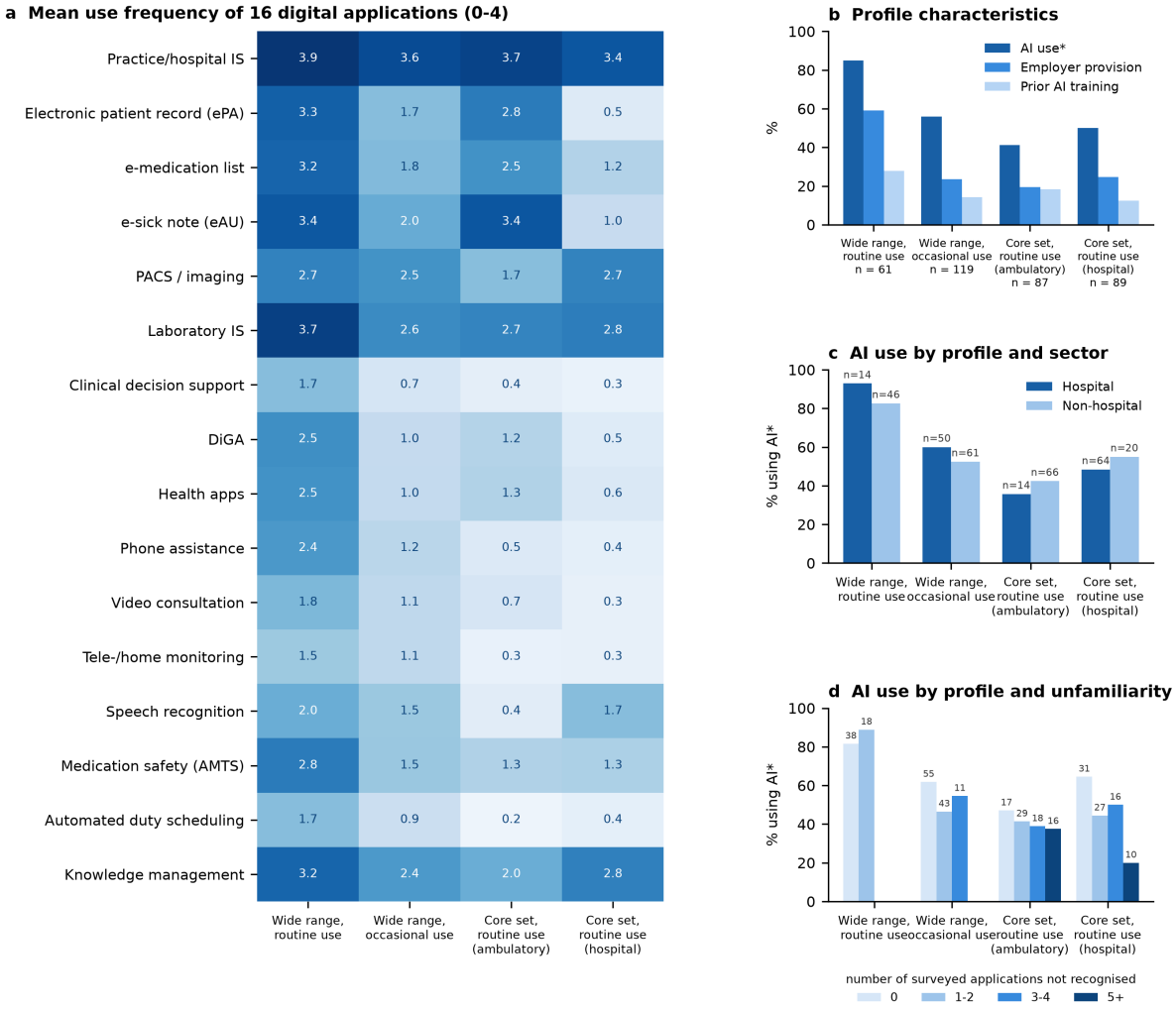
The four-class solution

Under the primary specification the fourth class partitions the core-set profile along the sector boundary (Supplementary Table 25, Supplementary Fig. 3): the two subclasses differ in which core applications they use routinely (ambulatory telematics versus imaging and speech recognition), not in AI use, employer provision or prior training. The partition is poorly identified (Supplementary Table 21) and is not preferred by the BIC.

Supplementary Table 25 | Four-class solution of the primary specification (sector partition of the core-set profile)

Class	n	AI use %	Employer provision %	Prior AI training %	Three-class membership (routine / occasional / core set)	Sector (hospital / ambulatory / other)	Mean posterior
Wide range, routine use	61	85.0	59.0	27.9	60 / 1 / 0	15 / 45 / 1	0.932
Wide range, occasional use	119	55.9	23.5	14.3	1 / 114 / 4	53 / 56 / 10	0.970
Core set, routine use (ambulatory profile)	87	41.2	19.5	18.4	7 / 7 / 73	14 / 73 / 0	0.951
Core set, routine use (hospital profile)	89	50.0	24.7	12.4	1 / 0 / 88	68 / 17 / 4	0.949

χ^2 across the four classes: AI use $\chi^2(3) = 28.8$, $p < 0.001$; provision $\chi^2(3) = 33.0$, $p < 0.001$; training $\chi^2(3) = 7.2$, $p = 0.067$. The two core-set subclasses do not differ in AI use (41.2% vs 50.0%, n = 80 and 84, $p = 0.26$), employer provision (19.5% vs 24.7%, $p = 0.41$) or prior training (18.4% vs 12.4%, $p = 0.27$).



Supplementary Fig. 3 | Four-class solution of the primary specification.

Mean use frequency of the 16 application items (0 = not available or not recognised to 4 = daily) and profile characteristics (AI use, employer provision, prior AI-specific training) across the four classes; the third and fourth classes partition the core-set profile by sector-typical availability patterns.

What is stable across specifications?

Supplementary Table 26 re-estimates the model under the three-level coding with 19 items and 20 starts, under the four-level coding with 19 and 17 items, and under the three-level coding with 16 items. The three use profiles are recovered in every specification, either as the three-class solution or as the first three classes of the four-class solution; the three-class partition under the 19-item, three-level specification agrees with the primary partition for 315 of 356 respondents (ARI 0.70). What varies is how the core-set profile is divided when a fourth class is added. Under the three-level coding, and under the four-level coding with 16 items, the fourth class separates hospital-typical from ambulatory-typical availability patterns; under the four-level coding with 17 items (tele-rounds included) it instead separates respondents who report the remaining applications as not available from respondents who do not recognise them (Supplementary Table 27). The two partitions lie close in likelihood, and which one a specification prefers depends on the item set. We therefore treat the heterogeneity within the core-set profile as two-dimensional and analyse each dimension directly:

sector by stratification and interaction tests (Supplementary Table 28), unfamiliarity as a person-level count (Supplementary Table 29; Supplementary Note 3, Supplementary Table 6).

Supplementary Table 26 | Stability of the three use profiles across codings and item sets

Specification	BIC-preferred k	Entropy (k = 3)	Class sizes, k = 3	ARI with primary (k = 3)	Respondents on the diagonal (of 356)	Class sizes, k = 4	ARI of k = 4 with primary
Primary: 16 items, four levels, 100 starts	3	0.902	69/122/165	1.000	356	61/119/87/89	0.626
19 items, three levels, 20 starts (submitted specification)	4	0.931	62/113/181	0.695	315	61/111/85/99	0.518
19 items, four levels, 100 starts	3	0.931	166/115/75	0.402	107	51/121/101/83	0.549
17 items, four levels, 100 starts	4	0.928	172/136/48	0.411	87	59/117/130/50	0.544
16 items, three levels, 100 starts	4	0.891	67/124/165	0.878	342	60/122/78/96	0.627

ARI, adjusted Rand index between modal partitions (1 = identical). Under the four-level coding with 19 or 17 items the three-class solution merges the two wide-range profiles and separates the core-set profile by familiarity (wide range / core set, others not available / core set, others unfamiliar), so its ARI with the primary partition is low; the three use profiles reappear as the first three classes of the corresponding four-class solutions (routine 51 and 59, occasional 121 and 117 respondents). For the 17-item specification the BIC-preferred four-class solution partitions the core-set profile by familiarity (Supplementary Table 27); for all other specifications the fourth class partitions it by sector.

Supplementary Table 27 | Two partitions of the core-set profile that compete across specifications

Partition	Class	n	AI use %	Employer provision %	Hospital %	Median 'unclear' answers	Median 'not available' answers
Familiarity (17 items, four levels, k = 4)	Wide range, routine use	59	84.5	61.0	23.7	0.0	1.0
Familiarity (17 items, four levels, k = 4)	Wide range, occasional use	117	52.7	21.4	41.9	1.0	1.0
Familiarity (17 items, four levels, k = 4)	Core set, others not available	130	52.4	24.6	53.1	1.0	7.0
Familiarity (17 items, four levels, k = 4)	Core set, others unfamiliar	50	37.2	20.0	36.0	5.5	4.0
Sector (16 items, four levels, k = 4)	Wide range, routine use	61	85.0	59.0	24.6	0.0	1.0
Sector (16 items, four levels, k = 4)	Wide range, occasional use	119	55.9	23.5	44.5	1.0	1.0
Sector (16 items, four levels, k = 4)	Core set, routine use (ambulatory profile)	87	41.2	19.5	16.1	2.0	4.0
Sector (16 items, four levels, k = 4)	Core set, routine use (hospital profile)	89	50.0	24.7	76.4	1.0	7.0

Identifiability: familiarity partition (17 items) reached by 26 of 100 starts with 10 distinct local maxima within 5 LL units, BIC 12872 vs 12892 for three classes, entropy 0.926; sector partition (16 items) reached by 2 of 100 starts with 21 distinct local maxima, BIC 12259 vs 12250, entropy 0.911. In the 17-item solution the two core-set subclasses differ in AI use by 15.2 percentage points.

Unanswered cells and competing solutions

Unanswered cells (20 of 5,696, in 16 respondents) were coded as not available. Excluding these respondents left the partition unchanged (n = 340, ARI 0.96). With unanswered cells treated as missing (StepMix categorical_nan), the BIC again selected three classes (12,177.6 versus 12,192.0 for four), the occasional-use profile was retained (113 of its 122 respondents), and AI use followed the same order across profiles (68.1%, 56.8% and 46.9%; $\chi^2(2) = 9.9$, $p = 0.007$), but 27 respondents of the core-set profile were assigned to the routine-use profile (101/119/136; ARI 0.68). This shift does not arise from the unanswered cells. Under the primary coding, a solution 0.65 log-likelihood units

below the maximum assigns 51 core-set respondents to the routine-use profile (125/118/113; ARI 0.57), lowering its AI use from 83.1% to 61.3% and employer provision from 56.5% to 37.6%, with the order of the three profiles unchanged. The occasional-use profile and the ordering of AI use and employer provision across profiles are therefore robust, whereas the boundary between the routine-use and core-set profiles, and with it the size and percentages of the routine-use profile, is weakly determined.

Script: analysis/lca_missing_sensitivity.py.

Sector

The profiles are sector-mixed, and the association between profile and AI use did not differ detectably between sectors (interaction p = 0.85); the association between profile and employer provision is confined to non-hospital respondents (interaction p = 0.019), consistent with the overlap between provision and practice ownership discussed in the main text.

Supplementary Table 28 | Profiles within hospital and non-hospital respondents (primary model)

Outcome	Stratum	Wide range, routine use	Wide range, occasional use	Core set, routine use	$\chi^2(2)$, p
AI use	hospital	81.2% (16)	59.2% (49)	48.1% (77)	6.3, 0.043
Employer provision	hospital	33.3% (18)	30.8% (52)	23.8% (80)	1.2, 0.562
Prior AI training	hospital	16.7% (18)	15.4% (52)	12.5% (80)	0.3, 0.844
AI use	non-hospital	83.7% (49)	51.6% (64)	43.8% (80)	20.6, < 0.001
Employer provision	non-hospital	64.7% (51)	20.0% (70)	17.6% (85)	38.7, < 0.001
Prior AI training	non-hospital	29.4% (51)	15.7% (70)	16.5% (85)	4.3, 0.114

Logistic regressions with profile and sector (hospital vs other) as predictors: AI use, profile likelihood-ratio $\chi^2(2) = 28.8$, $p < 0.001$, hospital OR 1.21 (0.77-1.92), profile-by-sector interaction $\chi^2(2) = 0.32$, $p = 0.85$ (n = 335); employer provision, profile $\chi^2(2) = 29.7$, $p < 0.001$, interaction $\chi^2(2) = 7.97$, $p = 0.019$; prior training, profile $\chi^2(2) = 3.7$, $p = 0.16$, interaction $p = 0.71$.

Unfamiliarity with the surveyed applications

Respondents did not recognise a median of one of the 16 applications (interquartile range 0 to 2); 58.7% did not recognise at least one and 9.6% five or more. The number of applications not recognised was unrelated to the number reported as not available (Spearman $\rho = 0.01$) and correlated with self-rated professional digital competence ($\rho = -0.26$, $p < 0.001$), whereas the number not available correlated only weakly with it ($\rho = -0.11$, $p = 0.031$). AI use declined with the number of applications not recognised, overall and within the core-set profile (Supplementary Table 29; trend p = 0.002; within the core-set profile OR 0.85 per application, 95% CI 0.74 to 0.99, adjusted for sector). Added to the multivariable model of Table 2, the number of applications not recognised was weakly associated with AI use (OR 0.87 per application, 95% CI 0.74–1.01), whereas the number not available and the mean use frequency among the remaining applications were not, and the effects of employer provision and private AI use were unchanged (Supplementary Note 3, Supplementary Table 6). With the number not recognised entered alongside profile and sector, its coefficient is attenuated (OR 0.91, 0.79 to 1.05), because most respondents with many unrecognised applications belong to the core-set profile. The variable counts applications the respondent did not recognise among those surveyed; it does not measure knowledge of artificial intelligence, which the questionnaire did not assess. Among the individual items, clinical decision support was not

recognised by 41.9% of respondents (41.4% in ambulatory care, 44.0% in hospitals); AI use was 47.5% among these respondents and 62.2% among the others (p = 0.007).

Supplementary Table 29 | AI use by the number of surveyed applications not recognised (16 items)

Applications not recognised	All respondents: AI use % (n)	Wide range, routine use	Wide range, occasional use	Core set, routine use
0	66.0 (141)	82.0 (39)	61.1 (54)	58.3 (48)
1-2	51.3 (117)	80.0 (20)	47.6 (42)	43.6 (55)
3-4	50.0 (48)	100.0 (4)	50.0 (14)	43.3 (30)
5+	37.9 (29)	100.0 (2)	66.7 (3)	29.2 (24)

Conclusion

Separating ‘not available’ from ‘unclear what this is’ and excluding three setting-bound items yields a three-class solution that the BIC prefers, that is reproducible and well classified, and that reproduces the three use profiles found under the 19-item, three-level specification. The fourth class of that specification reflected sector-typical availability patterns under a coding that treated unavailability and unfamiliarity as one state; under the separated coding it competes with a familiarity partition, and neither is robust to the item set. The substantive result is unchanged: professional AI use is close to one half in the core-set profile in both sectors, prior training is rare in all profiles, and AI use is lowest among physicians who do not recognise the surveyed applications.

Scripts: analysis/lca.py (Supplementary Tables 20 to 28, Fig. 2, Supplementary Fig. 3), analysis/item_screen.py (Supplementary Table 19), analysis/application_counts.py (Supplementary Table 29), analysis/note8_tables.py (assembles the tables).

Supplementary Note 9. CHERRIES checklist (Checklist for Reporting Results of Internet E-Surveys)

Reporting of this cross-sectional web-based survey follows the CHERRIES checklist. Items that could not be reported, or are not applicable, given the anonymous convenience-sample design are stated as such; these limitations are acknowledged in the manuscript.

Supplementary Table 30 | CHERRIES checklist (Checklist for Reporting Results of Internet E-Surveys)

Checklist item	CHERRIES description	Reported in this study (location)
1. Design		
Describe survey design	Describe target population, sample, whether the sample was self-selected, and the general design (e.g. cross-sectional).	Cross-sectional web-based survey; non-probability convenience sample of physicians practising in Germany (target population N ≈ 446,100). Self-selected respondents recruited via the Hartmannbund email distribution list. (Methods > Study design and participants; Recruitment.)
2. IRB (Institutional Review Board) approval and informed consent process		
IRB approval	Mention whether the study has been approved by an IRB/ethics committee.	Approved by University of Cambridge (Ref. 736, 15 May 2025) and Offenburg University of Applied Sciences (Ref. 04-25, 27 June 2025); conducted per the Declaration of Helsinki. (Methods > Ethics.)
Informed consent	Describe the informed consent process and where it was displayed.	A privacy notice and participant information sheet preceded the questionnaire; participation was conditional on informed consent from all participants. (Methods > Ethics.)
Data protection	State whether and how the survey conforms with data-protection regulations.	Data collected in compliance with the GDPR; participation was voluntary and anonymous, with no personally identifying data collected. (Methods > Ethics; Recruitment.)
3. Development and pre-testing		
Development and testing	State how the survey was developed, including pre-testing, usability and functionality.	Instrument newly developed (no validated instrument available); items drew on prior exploratory interviews and related projects. Piloted with 15 members of the working group of junior physicians (Arbeitskreis Junge Ärztinnen und Ärzte) of the Hartmannbund, that is, qualified doctors undergoing specialty training, who completed the full questionnaire and gave structured feedback on comprehensibility, terminology and response options; response categories were revised accordingly. Full instrument in Supplementary Note 7. (Methods > Instrument development.)
4. Recruitment process and description of the sample having access to the questionnaire		
Open survey vs closed survey	State whether the survey was open to any visitor, or a closed survey for a defined sample.	Closed survey: distributed to the defined membership of the Hartmannbund email distribution list; not publicly advertised or indexed. (Methods > Recruitment and participants.)
Contact mode	Indicate whether initial contact with potential participants was made on the Internet.	Initial contact made electronically by email via the central Hartmannbund distribution list. (Methods > Recruitment and participants.)
Advertising the survey	How/where was the survey announced or advertised?	Announced through a single email invitation (December 2025) with one email reminder (January 2026); no other advertising channels. (Methods > Recruitment and participants.)
5. Survey administration		
Web / E-mail	State whether it was a Web- or email-based survey.	Web-based survey administered via Microsoft Forms; the link was distributed by email. (Methods – Study design and participants.)
Context	Describe the Web site/page in which the survey was hosted.	Hosted on Microsoft Forms, optimised for desktop and mobile; the survey formed part of a larger project on digitalisation and AI adoption among physicians. (Methods > Study design and participants.)
Mandatory / voluntary	Was the survey mandatory or voluntary?	Voluntary and anonymous. (Methods > Recruitment and participants.)

Incentives	Were any incentives offered?	No incentives were offered. (Methods > Recruitment and participants.)
Time / Date	In which time frame was the survey conducted?	December 2025 to February 2026. (Methods > Study design and participants.)
Randomisation of items or questionnaires	To prevent bias, were items/pages randomised?	Not applicable. Items were presented in a fixed order; no randomisation was used.
Adaptive questioning	Were items dynamically displayed based on prior answers, to reduce complexity?	Conditional branching was used for eight items: specialty and professional experience (items 8 and 9, shown to physicians; items 10 and 11, shown to other professional groups), the form of employer-provided AI (item 15, shown to respondents reporting provision), areas of AI use (item 20, shown to respondents reporting use or uncertainty about use) and the type and usefulness of prior training (items 23 and 24, shown to respondents reporting training). All other items were displayed to all respondents.
Number of items	What was the number of questionnaire items per page and in total?	33 items: two consent items, nine socio-demographic items and 22 items in the three thematic blocks described in the Methods (digitalisation and AI status quo; unsanctioned AI use, analysed elsewhere; training experience and preferences). Full questionnaire in Supplementary Note 7.
Number of screens (pages)	Over how many pages was the questionnaire distributed?	Number of screens/pages not reported. Administered via Microsoft Forms; average completion time about 8 to 10 minutes after piloting. (Methods > Instrument development.)
Completeness check	Was completeness checked before submission?	Items marked as required in Supplementary Note 7 could not be skipped (enforced by Microsoft Forms). The matrix item on digital applications (item 13) was not required; missing values there were handled in the analysis (at least 15 of 19 valid items for the latent class analysis, Supplementary Note 8).
Review step	Could respondents review and change their answers (e.g. Back button)?	Not explicitly reported. Microsoft Forms permits back-navigation before final submission.
6. Response rates		
Unique site visitor	How were unique visitors determined (cookie, IP, log file)?	Not determinable. Anonymous email-list distribution with no site-visitor tracking; number of invitations delivered is not quantifiable. Acknowledged as a limitation. (Discussion.)
View rate	Ratio of unique survey visitors / unique site visitors.	Not calculable (denominator unknown; self-selected sample without a quantifiable response rate). (Discussion.)
Participation rate	Ratio of unique visitors who agreed to participate / unique first survey-page visitors.	Not calculable for the same reason.
Completion rate	Ratio of users who finished the survey / users who agreed to participate.	403 questionnaires were received; 41 excluded on predefined eligibility/data criteria, yielding 362 analysed. A completion rate relative to those who started cannot be computed. (Results > Participant characteristics; Supplementary Note 1.)
7. Preventing multiple entries from the same individual		
Cookies used	Were cookies used to assign a unique user identifier?	Not reported; the survey was anonymous.
IP check	Were IP addresses checked to prevent duplicates?	IP addresses were not collected (anonymous, GDPR-compliant); no IP-based duplicate check.
Log file analysis	Were other techniques used to analyse the log file for identical entries?	Raw responses were screened for duplicates during data cleaning; no duplicate entries were identified. (Supplementary Note 1.)
Registration	Was the survey open only to registered users identified by a code/password?	No registration or login; open to any recipient of the email invitation but confined to that closed distribution list.
8. Analysis		
Handling of incomplete questionnaires	Were incomplete questionnaires analysed?	Records missing key demographic variables were excluded (Supplementary Note 1); the regression used complete cases (n = 328). 21 participants with undefined AI-use status were excluded from the comparisons across the four adoption groups (n = 341); the latent class analysis required at least 15 of 19 valid application items (n = 356). (Methods > Measures/Statistical analysis.)
Questionnaires submitted with an atypical timestamp	Were surveys with atypical completion times identified/excluded?	No timestamp-based exclusion was applied or reported.

Statistical correction	Were weighting or propensity-score methods used to adjust for a non-representative sample?	No post-stratification weighting applied; representativeness was assessed by comparison with Bundesärztekammer statistics (Supplementary Note 2). Multiplicity controlled by Benjamini–Hochberg FDR correction; robustness checked in sensitivity analyses (Supplementary Note 3). (Methods > Statistical analysis.)

Supplementary Note 10. Adoption groups in detail: the shape of group differences, the two non-user groups, preferences by exposure, early-career physicians, the form of employer-provided AI, providers by sector, and respondents unsure of their AI use

Trend and decomposition (Supplementary Table 31)

For each of the 30 tested options, a logistic regression with the ordered group (0 = no current intention, 1 = interested non-user, 2 = non-routine use, 3 = routine use) as a linear term, adjusted for age (four categories), sex, self-rated professional digital competence and sector (complete cases, n = 328; Model A), gives the odds ratio per group; a likelihood-ratio test compares this model with one in which the four groups are free (2 df) and indicates departure from linearity. Model F adds prior AI-specific training, private AI use and employer provision (the covariate set of Table 2). The same covariates as in Model A are used for the contrast between non-users without intention and all other respondents and for three contrasts that decompose the trend: interested versus no intention among non-users (n = 143), users versus non-users, and routine versus non-routine use among users (n = 185); among users, the number of coded areas of AI use (0–5) was added at a given frequency. Options are classified by rule: 'gradient' if the linear term is significant after Benjamini–Hochberg correction, there is no departure from linearity and the routine-versus-non-routine contrast is significant; 'threshold' if the contrast between non-users without intention and all others is significant after correction and the option is not a gradient; otherwise 'invariant'. Two options are gradients (advanced and strategic level), three are thresholds (large language models, ethics and law, diagnostic systems), one combines a threshold with a rise among routine users (commercial AI vendors), and 24 are invariant.

Supplementary Table 31 | Trend across the four ordered adoption groups and its decomposition, all 30 tested options (odds ratios with 95% CI)

Option	OR per group, Model A (95% CI)	BH p	Non-linearity p	OR per group, Model F (95% CI)	No intention vs rest, OR (95% CI), BH p	Interested vs no intention (non-users), OR (95% CI)	Routine vs non-routine (users), OR (95% CI)	Per area of use (users), OR (95% CI)	Pattern
Self-paced online courses	1.23 (0.97-1.58)	0.253	0.780	1.25 (0.92-1.68)	0.58 (0.31-1.10), 0.202	1.58 (0.72-3.44)	1.34 (0.64-2.82)	1.27 (0.83-1.95)	invariant
Live webinars	1.25 (1.02-1.54)	0.132	0.182	1.19 (0.93-1.53)	0.49 (0.27-0.88), 0.085	2.16 (1.07-4.37)	1.55 (0.83-2.88)	1.34 (0.95-1.88)	invariant
Nuggets (15-30 min)	0.92 (0.75-1.14)	0.606	0.341	0.83 (0.64-1.08)	0.98 (0.54-1.78), 0.960	1.33 (0.65-2.72)	1.11 (0.59-2.08)	1.18 (0.86-1.63)	invariant
Blended learning	1.09 (0.87-1.35)	0.606	0.091	1.18 (0.90-1.54)	0.63 (0.33-1.19), 0.262	2.05 (0.97-4.36)	1.47 (0.77-2.84)	1.00 (0.71-1.40)	invariant
On-site sessions	1.08 (0.87-1.35)	0.606	0.439	1.09 (0.83-1.42)	0.75 (0.40-1.39), 0.493	1.44 (0.69-3.01)	1.50 (0.78-2.91)	0.68 (0.47-0.98)	invariant
Short formats (1-2 h)	1.21 (0.92-1.58)	0.390	0.925	1.08 (0.78-1.49)	0.69 (0.35-1.38), 0.422	1.34 (0.58-3.07)	1.44 (0.63-3.29)	1.50 (0.91-2.46)	invariant
Half-day	1.05 (0.82-1.34)	0.816	0.624	1.01 (0.76-1.35)	0.77 (0.39-1.54), 0.601	1.29 (0.56-2.98)	0.81 (0.40-1.62)	0.83 (0.56-1.22)	invariant
Longitudinal	1.16 (0.88-1.52)	0.528	0.061	1.31 (0.95-1.80)	0.45 (0.19-1.08), 0.185	3.25 (1.17-9.03)	1.64 (0.72-3.71)	0.98 (0.65-1.47)	invariant
Full-day	1.22 (0.87-1.70)	0.508	0.559	1.53 (1.03-2.29)	0.46 (0.15-1.38), 0.264	1.97 (0.55-7.02)	0.90 (0.36-2.21)	0.95 (0.58-1.56)	invariant
Application-oriented	0.94 (0.65-1.36)	0.826	0.088	0.90 (0.57-1.42)	0.75 (0.28-1.99), 0.652	1.38 (0.41-4.69)	0.30 (0.08-1.12)	1.35 (0.79-2.31)	invariant
Introductory	1.14 (0.93-1.41)	0.459	0.766	1.19 (0.92-1.55)	0.86 (0.48-1.55), 0.693	0.89 (0.44-1.79)	1.14 (0.61-2.15)	1.15 (0.82-1.61)	invariant
Advanced / specialised	1.89 (1.48-2.42)	< 0.001	0.648	2.00 (1.49-2.68)	0.28 (0.13-0.59), 0.008	1.68 (0.69-4.09)	2.48 (1.30-4.75)	1.73 (1.19-2.52)	gradient
Strategic	1.52 (1.20-1.92)	0.009	0.474	1.46 (1.10-1.93)	0.48 (0.23-0.97), 0.125	1.37 (0.57-3.30)	2.22 (1.17-4.21)	1.92 (1.33-2.76)	gradient

Option	OR per group, Model A (95% CI)	BH p	Non-linearity p	OR per group, Model F (95% CI)	No intention vs rest, OR (95% CI), BH p	Interested vs no intention (non-users), OR (95% CI)	Routine vs non-routine (users), OR (95% CI)	Per area of use (users), OR (95% CI)	Pattern
Diagnostic systems	1.23 (1.00-1.52)	0.162	0.042	1.12 (0.87-1.45)	0.40 (0.22-0.74), 0.022	3.04 (1.43-6.45)	1.22 (0.67-2.23)	1.38 (0.99-1.91)	threshold
Treatment planning	1.06 (0.86-1.30)	0.727	0.027	1.02 (0.79-1.31)	0.52 (0.28-0.96), 0.125	2.49 (1.17-5.28)	0.62 (0.34-1.15)	1.38 (1.00-1.91)	invariant
Practical implementation	1.00 (0.81-1.23)	0.992	0.091	0.96 (0.75-1.24)	0.63 (0.35-1.16), 0.262	2.08 (1.01-4.28)	0.74 (0.40-1.35)	0.92 (0.67-1.27)	invariant
Large language models	1.43 (1.14-1.79)	0.017	0.067	1.37 (1.05-1.79)	0.28 (0.13-0.58), 0.008	2.83 (1.22-6.56)	0.83 (0.45-1.53)	1.06 (0.77-1.44)	threshold
Data protection	0.99 (0.78-1.24)	0.931	0.188	1.17 (0.88-1.55)	1.52 (0.82-2.80), 0.275	0.49 (0.22-1.09)	1.28 (0.65-2.51)	0.93 (0.65-1.33)	invariant
Diagnostic imaging	1.40 (1.09-1.79)	0.056	0.232	1.23 (0.91-1.66)	0.48 (0.22-1.03), 0.166	1.61 (0.65-4.00)	2.36 (1.15-4.84)	1.35 (0.95-1.93)	invariant
Critical appraisal	0.91 (0.70-1.18)	0.606	0.029	0.95 (0.69-1.31)	2.10 (1.09-4.04), 0.114	0.30 (0.12-0.76)	1.11 (0.52-2.37)	0.64 (0.41-1.01)	invariant
Ethics and law	0.70 (0.53-0.94)	0.080	0.028	0.79 (0.56-1.11)	3.61 (1.81-7.19), 0.008	0.29 (0.12-0.72)	1.54 (0.59-4.01)	1.12 (0.69-1.81)	threshold
Patient communication	0.90 (0.67-1.21)	0.606	0.249	0.84 (0.59-1.21)	0.76 (0.32-1.81), 0.652	2.03 (0.77-5.40)	0.82 (0.33-2.05)	0.67 (0.38-1.17)	invariant
Chambers of physicians	0.98 (0.79-1.22)	0.911	0.245	0.93 (0.71-1.21)	0.98 (0.54-1.81), 0.960	1.42 (0.66-3.05)	1.47 (0.79-2.75)	1.05 (0.76-1.46)	invariant
Professional associations	1.09 (0.88-1.35)	0.606	0.299	1.13 (0.88-1.46)	0.62 (0.34-1.12), 0.224	1.67 (0.83-3.39)	0.83 (0.44-1.55)	1.03 (0.74-1.42)	invariant
Employers	1.13 (0.87-1.48)	0.606	0.930	1.11 (0.81-1.53)	0.79 (0.37-1.69), 0.652	1.10 (0.41-2.91)	1.16 (0.57-2.39)	1.55 (1.05-2.27)	invariant
Specialty societies	1.19 (0.97-1.47)	0.253	0.283	1.17 (0.91-1.51)	0.54 (0.30-0.98), 0.125	1.76 (0.83-3.72)	0.89 (0.49-1.62)	1.00 (0.73-1.37)	invariant
Assoc. of statutory health insurance	0.90 (0.71-1.14)	0.606	0.700	0.83 (0.62-1.12)	1.14 (0.60-2.19), 0.740	1.14 (0.53-2.47)	1.13 (0.57-2.27)	1.39 (0.96-1.99)	invariant
Universities	1.24 (0.98-1.58)	0.228	0.222	1.30 (0.98-1.74)	0.60 (0.30-1.22), 0.262	1.57 (0.67-3.72)	2.03 (1.01-4.08)	1.28 (0.91-1.81)	invariant
Commercial AI vendors	1.50 (1.14-1.98)	0.032	0.014	1.63 (1.16-2.28)	0.16 (0.05-0.55), 0.022	6.85 (1.83-25.63)	2.28 (1.05-4.95)	1.15 (0.79-1.66)	threshold + routine
Private providers	1.42 (1.03-1.95)	0.132	0.300	1.65 (1.13-2.41)	0.38 (0.13-1.14), 0.195	2.62 (0.71-9.63)	2.48 (1.01-6.08)	1.12 (0.75-1.68)	invariant

Composition of the level and topic items (Supplementary Table 32)

The number of levels named rose across the groups (1.75, 1.94, 2.18, 2.46; Kruskal–Wallis $p < 0.001$), the share naming only introductory content fell (38%, 34%, 24%, 20%) and the share naming advanced without introductory content rose (10%, 13%, 16%, 31%). The topic item allowed three selections; 70 to 89% of each group used all three, so topic percentages are relative priorities. Non-users without intention named fewer topics (2.32 against 2.73 to 2.85) and more often none (14.3% against 0 to 1.2%); the share of governance topics among their selections was 42%, against 17 to 23% in the other groups. When the number of levels named is held constant, the gradient for advanced training persists (OR 1.45 per group, $p = 0.001$) and the gradient for strategic training does not (OR 1.03, $p = 0.83$).

Supplementary Table 32 | Composition of the level and topic items by adoption group

Group	n	Levels named (mean of 4)	Introductory only (no advanced or strategic), %	Introductory and advanced, %	Advanced without introductory, %	Topics named (mean, max. 3)	Three topics named, %	Any governance topic, %	Governance share of named topics, %
No current intention	63	1.75	38.1	7.9	9.5	2.32	69.8	60.3	42.3
Interested non-users	86	1.94	33.7	10.5	12.8	2.78	82.6	34.9	17.3
Non-routine use	85	2.18	23.5	20.0	16.5	2.73	82.4	44.7	22.6
Routine use	107	2.46	19.6	29.9	30.8	2.85	88.8	48.6	22.3

Supplementary Table 33 | Composite content measures and exposure characteristics by adoption group

Group	n	Any governance topic, %	Any clinical application topic, %	No topic, %	Private AI use, %	Informs actively about AI, %	Passive engagement, %	Areas of AI use, median (mean of 5; users only)
No current intention	63	60.3	49.2	14.3	34.9	28.6	25.4	–
Interested non-users	86	34.9	77.9	1.2	52.3	31.4	19.8	–
Non-routine use	85	44.7	76.5	1.2	72.9	32.9	7.1	1 (1.33)
Routine use	107	48.6	73.8	0.0	79.4	65.4	7.5	2 (2.10)

Any governance topic: data protection, ethics and law or critical appraisal; any clinical application topic: diagnostic systems, treatment planning or diagnostic imaging; private AI use, active information-seeking and passive engagement from item 21; areas of AI use among users from item 20). Fisher's exact tests for adjacent groups: any governance topic, interested non-users versus no intention $p = 0.003$; any clinical topic $p < 0.001$; private AI use, non-routine users versus interested non-users $p = 0.007$; passive engagement $p = 0.023$; informs actively, routine versus non-routine users $p < 0.001$; areas of use, routine versus non-routine users, Mann–Whitney $p < 0.001$.

Topic structure

A principal component analysis of the ten topic indicators, on Pearson and on tetrachoric correlations (to allow for the selection cap), yields two components above chance in a parallel analysis: governance (critical appraisal, data protection, ethics and law) versus clinical application (diagnostic systems, treatment planning, imaging), and workflow (practical implementation, patient communication) versus technical diagnostics (imaging, diagnostic systems); loadings agree between the two methods ($\rho = 0.99$). On the first component, non-users without intention lay 0.7 to 1.0 standard units from each other group after adjustment for age, sex, competence and sector (adjusted $p = 0.001$ or smaller), and the other three groups did not differ ($p = 0.09$ to 0.79). On the second component, routine users tended towards technical diagnostics and the two middle groups (interested non-users and non-routine users) towards workflow topics (Kruskal–Wallis $p = 0.026$; adjusted group test $p = 0.07$); this is reported as a hypothesis.

The two non-user groups (Supplementary Tables 34 to 36)

The two groups are compared on all preference options in Supplementary Note 6 (Supplementary Tables 17 and 18). Supplementary Table 34 compares their composition; Supplementary Table 35 the closed barrier item (six predefined reasons, up to three selections); Supplementary Table 36 on the free-text reasons, coded against ten categories defined before coding and reviewed independently by two authors. Non-users without intention were not older, more experienced or more often practice-based, and rated their digital competence similarly; they used AI privately less often (34.9% versus 52.3%), almost none had employer provision (4.8% versus 9.3%), and more did not recognise three or more of the surveyed applications (34.9% versus 23.3%). In free text, 39.7% of the group without intention gave an evaluative reason (no perceived benefit, doubts about reliability or principled rejection) against 7.0% of interested non-users, whereas not knowing where to start was written only by interested non-users (14.0% versus 0%). The word 'trust' (Vertrauen) appeared in one free-text answer among the 149 non-users.

Supplementary Table 34 | Composition of the two non-user groups

Characteristic	No current intention (n = 63)	Interested non-users (n = 86)	Users (n = 192)
age >= 50 years, %	49.2	50.0	44.3
age 60+, %	23.8	23.3	22.4
age < 40, %	42.9	36.0	34.4
male, %	50.8	43.0	55.2
hospital, %	46.0	39.5	42.2
ambulatory, %	49.2	57.0	54.2
other sector, %	4.8	3.5	3.6
> 25 years experience, %	44.4	40.7	37.0
self-rated competence, mean	3.2	3.0	3.5
private AI use, %	34.9	52.3	76.6
employer provides AI, %	4.8	9.3	47.9
prior AI training, %	7.9	17.4	21.4
applications not recognised, mean (16 items)	2.2	1.6	1.2
not recognised >= 3, %	34.9	23.3	18.2

Supplementary Table 35 | Closed barrier item by non-user group

Barrier (closed item, up to three selections)	Interested non-users, n (%; 95% CI)	No current intention, n (%; 95% CI)	Difference, pp (Newcombe 95% CI)	Fisher p	BH-adjusted p (6 tests)
Not available	55 (64.0; 53.4-73.3)	36 (57.1; 44.9-68.6)	-6.8 (-22.2 to 8.8)	0.496	0.496
Lack of knowledge / training	42 (48.8; 38.6-59.2)	21 (33.3; 22.9-45.6)	-15.5 (-30.2 to 0.5)	0.066	0.199
Data protection / security concerns	28 (32.6; 23.6-43.0)	24 (38.1; 27.1-50.4)	+5.5 (-9.6 to 20.8)	0.492	0.496
Insufficient system integration	23 (26.7; 18.5-36.9)	12 (19.0; 11.2-30.4)	-7.7 (-20.5 to 6.3)	0.330	0.496
No perceived benefit	12 (14.0; 8.2-22.8)	19 (30.2; 20.2-42.4)	+16.2 (2.9 to 29.7)	0.024	0.143
Lack of time / capacity	19 (22.1; 14.6-31.9)	10 (15.9; 8.9-26.8)	-6.2 (-18.3 to 7.0)	0.406	0.496

Supplementary Table 36 | Free-text reasons for not pursuing AI-related training, by non-user group (categories defined before coding; several codes per answer; coding reviewed independently by two authors)

Category (free text; several codes per answer possible)	Interested non-users (n = 86), n (%)	No current intention (n = 63), n (%)	Difference, pp (Newcombe 95% CI)	Fisher p (uncorrected)
Lack of time or capacity	36 (41.9)	23 (36.5)	-5.4 (-20.4 to 10.4)	0.611
No access: no offers, no systems or support at work, costs, infrastructure	31 (36.0)	18 (28.6)	-7.5 (-21.8 to 7.8)	0.380
No perceived benefit, need or relevance for own work	4 (4.7)	16 (25.4)	+20.7 (9.4 to 33.0)	< 0.001
Doubts about the maturity, error rate or controllability of AI; unwillingness to rely on its output	2 (2.3)	9 (14.3)	+12.0 (3.2 to 22.8)	0.009
Principled rejection of AI in medicine; scepticism about the 'hype'; preference for established routine	1 (1.2)	8 (12.7)	+11.5 (3.5 to 22.0)	0.005
Data protection, legal, ethical or environmental concerns	2 (2.3)	2 (3.2)	+0.8 (-5.4 to 8.7)	1.000
Career stage: approaching retirement, other training priorities	3 (3.5)	3 (4.8)	+1.3 (-5.7 to 9.9)	0.698
Difficulty judging the seriousness of providers and offers	1 (1.2)	2 (3.2)	+2.0 (-3.6 to 9.8)	0.574
Lack of basic knowledge or orientation: not knowing where or how to start, which training is adequate, entry hurdle	12 (14.0)	0 (0.0)	-14.0 (-22.8 to -5.8)	0.001

Category (free text; several codes per answer possible)	Interested non-users (n = 86), n (%)	No current intention (n = 63), n (%)	Difference, pp (Newcombe 95% CI)	Fisher p (uncorrected)
No reason given, not codable, or 'nothing keeps me / I already train'	13 (15.1)	5 (7.9)	-7.2 (-17.3 to 4.0)	0.212
Any evaluative reason (no perceived benefit, reliability doubts or principled rejection)	6 (7.0)	25 (39.7)	+32.7 (19.3 to 45.6)	< 0.001

Preferences by exposure (Supplementary Tables 37-39)

Two markers of exposure to AI outside employer-provided systems were related to preferences among respondents with defined use status: private AI use (n = 214 of 341) and having ever used AI at work without formal authorisation (first response option of item 32; n = 90 of 314 with a valid answer; 87% of them used AI privately; 22 of the 90 reported no current professional AI use). Model S1 adjusts for use status, age, sex, competence and sector; model S2 additionally for the other exposure marker, prior training and employer provision. Those reporting unauthorised use named any governance topic in 34.4% of cases against 49.6% (S1 OR 0.53, 95% CI 0.30 to 0.92; S2 0.65, 0.36 to 1.16) and any clinical application topic in 80.0% against 67.9%. Private AI use was associated with naming any governance topic in 41.1% against 55.1% (S1 OR 0.50, 0.31 to 0.82; unchanged in S2) and with lower endorsement of ethics and law (12.6% versus 22.0%) and data protection (24.3% versus 35.4%). No single option differed after correction across 30 options in the adjusted models. Among physicians who neither used AI privately nor reported unauthorised use, 57% named a governance topic; among those who did one of the two, 43% or 33%; among those who did both, 35%. The item on unauthorised use is analysed in a separate manuscript and is used here only as a marker of exposure.

Supplementary Table 37 | Composite content measures, ethics and law, and data protection by self-reported unauthorised use (n = 314 with defined use status and a valid answer)

Option	Unauthorised use reported, % (n)	Not reported, % (n)	Fisher p	BH-adjusted p (30)	Model S1: OR (95% CI)	BH-adjusted p (S1)	Model S2: OR (95% CI)
Data protection	18.9 (17)	30.8 (69)	0.036	0.154	0.55 (0.29–1.05)	0.328	0.65 (0.33–1.30)
Ethics and law	8.9 (8)	17.9 (40)	0.056	0.167	0.39 (0.17–0.92)	0.231	0.48 (0.20–1.17)
Any governance topic	34.4 (31)	49.6 (111)	0.017	-	0.53 (0.30–0.92)	-	0.65 (0.36–1.16)
Any clinical application	80.0 (72)	67.9 (152)	0.038	-	1.56 (0.82–2.95)	-	1.49 (0.79–2.91)

Note: –, not applicable: composite measures are not part of the 30-option family and were not corrected for multiple testing.

Supplementary Table 38 | Composite content measures, ethics and law, and data protection by private AI use (n = 341 with defined use status)

Option	Private AI use, % (n)	No private AI use, % (n)	Fisher p	BH-adjusted p (30)	Model S1: OR (95% CI)	BH-adjusted p (S1)	Model S2: OR (95% CI)
Data protection	24.3 (52)	35.4 (45)	0.035	0.115	0.50 (0.29–0.87)	0.091	0.45 (0.25–0.82)
Ethics and law	12.6 (27)	22.0 (28)	0.032	0.115	0.41 (0.21–0.80)	0.087	0.50 (0.24–1.05)
Any governance topic	41.1 (88)	55.1 (70)	0.014	-	0.50 (0.31–0.82)	-	0.50 (0.29–0.86)
Any clinical application	75.7 (162)	63.0 (80)	0.014	-	1.62 (0.95–2.75)	-	1.67 (0.93–2.98)

Note: –, not applicable: composite measures are not part of the 30-option family and were not corrected for multiple testing.

Supplementary Table 39 | Governance and clinical application topics by the two exposure markers combined

Unauthorised use reported	Private AI use	n	Any governance topic, %	Any clinical application topic, %	Professional AI use, %
no	no	107	57.0	59.8	29.0
no	yes	117	42.7	75.2	63.2
yes	no	12	33.3	91.7	75.0
yes	yes	78	34.6	78.2	75.6

Early-career physicians (Supplementary Tables 40 and 41)

Whether a respondent was in specialty training was not recorded; professional experience counted from licensure serves as a proxy. Sixty-five respondents (18.0%) reported up to four years, 98.5% of them under 40 and 86% hospital-based; age and experience are nearly collinear (Spearman ρ = 0.91), so the adjusted models (sex, digital competence, sector, use status) do not include age. Early-career physicians did not differ detectably from their colleagues in AI use, training or employer provision, used AI privately somewhat more often and reported unauthorised use twice as often (Supplementary Table 40). Four provider options differed in unadjusted Fisher tests after correction (employers, universities, professional associations, associations of statutory health insurance); none remained after adjustment for sector and the other covariates, and none differed when the comparison was restricted to physicians under 40 (64 versus 66 respondents; Supplementary Table 41 and analysis/early_career.py).

Supplementary Table 40 | AI use, training and access by professional experience

Measure	0–4 years since licensure, % (n)	5 years or more, % (n)	Difference, pp (Newcombe 95% CI)	Fisher p
Professional AI use (defined status)	60.0 (60)	55.6 (279)	+4.4 (-9.4 to 17.3)	0.568
Prior AI-specific training	13.8 (65)	18.0 (295)	-4.1 (-12.1 to 7.0)	0.474
Employer provides AI	27.7 (65)	29.5 (295)	-1.8 (-12.7 to 11.1)	0.880
Private AI use	70.8 (65)	60.0 (295)	+10.8 (-2.4 to 22.0)	0.121
Self-reported unauthorised use (valid Q32)	50.0 (62)	23.9 (272)	+26.1 (12.9 to 39.1)	< 0.001

Supplementary Table 41 | The four provider options that differed by professional experience in unadjusted comparisons: unadjusted and adjusted (BH correction across the 30 tested options)

Option	0–4 years (n = 65), % (95% CI)	≥5 years (n = 295), % (95% CI)	Difference, pp (Newcombe 95% CI)	Fisher, BH-adjusted p	Adjusted OR (95% CI)	Adjusted, BH-adjusted p
Professional associations	40.0 (29.0-52.1)	60.3 (54.7-65.8)	-20.3 (-32.6 to -6.9)	0.035	0.54 (0.28–1.03)	0.674
Employers	81.5 (70.4-89.1)	43.1 (37.5-48.8)	+38.5 (26.0 to 47.9)	< 0.001	1.74 (0.74–4.08)	0.674
Assoc. of SHI physicians	20.0 (12.1-31.3)	38.0 (32.6-43.6)	-18.0 (-27.7 to -5.5)	0.045	1.21 (0.53–2.78)	0.801
Universities	46.2 (34.6-58.1)	24.4 (19.9-29.6)	+21.7 (9.1 to 34.6)	0.011	2.14 (1.09–4.20)	0.674

Form of employer-provided AI and areas of use (Supplementary Table 42)

Of the 105 respondents with employer provision, 45.7% described the AI as integrated into existing medical software, 33.3% as standalone tools or programs such as chatbots or speech recognition, 13.3% as pilots of external software and 1.9% as in-house models; free-text entries named a CT application, an intranet tool based on ChatGPT, Amboss, text summarisation in Acrobat Reader, PACS-integrated AI and office tools. Among users with provision, those with standalone tools named administration more often (77.4% versus 54.1%, p = 0.041); the other areas did not differ. Among the 84 users without provision, 76.2% used AI privately, 50.0% of those with a valid answer reported unauthorised use, and free-text entries named ChatGPT, OpenEvidence and private accounts. Across all users, 57.3% named diagnostics or treatment planning.

Supplementary Table 42 | Form of employer-provided AI

Form of employer-provided AI (item 15)	Respondents with provision (n = 105), n (%)	Users with provision (n = 92), n (%)
Standalone tools or programs (e.g. AI models or chatbots, speech recognition)	35 (33.3)	31 (33.7)
Integrated into existing medical software (e.g. diagnostic systems, decision support)	48 (45.7)	44 (47.8)
In trial / pilot phase (external software)	14 (13.3)	13 (14.1)
In-house developed AI models	2 (1.9)	1 (1.1)
No AI tools available	0 (0.0)	0 (0.0)
Don’t know	0 (0.0)	0 (0.0)
Other (free text)	6 (5.7)	3 (3.3)

Preferred providers by sector (Supplementary Table 43)

Employers were named as a preferred provider by 82.2% of hospital respondents and 24.1% of ambulatory respondents; associations of statutory health insurance physicians showed the reverse pattern; chambers of physicians and specialty societies were named without detectable differences between the sectors.

Supplementary Table 43 | Preferred providers of AI-related CME by sector (χ^2 tests, BH correction across the eight provider options)

Provider	Hospital (n = 152), % (95% CI)	Ambulatory care (n = 195), % (95% CI)	Difference hospital – ambulatory, pp (Newcombe 95% CI)	BH-adjusted p (8 tests)
Chambers of physicians	61.8 (53.9-69.2)	68.7 (61.9-74.8)	-6.9 (-16.9 to 3.1)	0.241
Professional associations	46.1 (38.3-54.0)	63.1 (56.1-69.5)	-17.0 (-27.1 to -6.5)	0.003
Employers	82.2 (75.4-87.5)	24.1 (18.6-30.6)	+58.1 (48.7 to 65.7)	< 0.001
Specialty societies	49.3 (41.5-57.2)	45.1 (38.3-52.1)	+4.2 (-6.3 to 14.6)	0.497
Assoc. of SHI physicians	14.5 (9.8-20.9)	50.8 (43.8-57.7)	-36.3 (-44.7 to -26.8)	< 0.001
Universities	40.1 (32.7-48.1)	19.5 (14.5-25.6)	+20.6 (11.0 to 30.0)	< 0.001
Commercial AI vendors	15.1 (10.3-21.7)	23.1 (17.7-29.5)	-7.9 (-16.0 to 0.5)	0.103

Provider	Hospital (n = 152), % (95% CI)	Ambulatory care (n = 195), % (95% CI)	Difference hospital – ambulatory, pp (Newcombe 95% CI)	BH-adjusted p (8 tests)
Private providers	13.2 (8.7-19.5)	12.8 (8.8-18.2)	+0.3 (-6.7 to 7.8)	0.926

Respondents unsure whether they use AI (Supplementary Table 44)

Fifteen respondents answered ‘I don’t know / am unsure’ to the item on professional AI use. They rated their professional digital competence lower than the sample, did not recognise more of the 16 surveyed applications, and nine did not know whether their employer provided AI. Seven of them nonetheless named areas in which they used AI applications (documentation, diagnostics, treatment planning, research); one wrote of using ChatGPT for DRG coding and for looking things up, Amboss, and Google Translate. Seven named the introductory level, 13 chambers of physicians and none commercial vendors.

Supplementary Table 44 | The 15 respondents unsure whether they use AI, compared with the whole sample

Measure	Unsure whether they use AI (n = 15)	All respondents (n = 362)
self-rated professional digital competence, mean (1-5)	2.47	3.28
surveyed applications not recognised, mean of 16	3.07	1.56
three or more applications not recognised, n	7	86
employer provision: ‘don’t know’, n	9	50
employer provision: yes, n	1	105
private AI use, n	7	225
prior AI-specific training, n	1	62
introductory level named, n	7	190
chambers of physicians named as provider, n	13	238
commercial AI vendors named as provider, n	0	69
hospital-based, n	7	152
age 50 years or older, n	7	171

Scripts: analysis/adoption_groups.py (Supplementary Tables 31,32), analysis/content_measures.py (Supplementary Table 33), analysis/topic_structure.py (topic structure), analysis/no_intention_group.py and analysis/nonuser_reasons.py with q30_author_review.csv (Supplementary Tables 34 to 36), analysis/exposure_markers.py (Supplementary Tables 37 to 39), analysis/early_career.py (Supplementary Tables 40 and 41), analysis/provision_form.py (Supplementary Table 42), analysis/providers_by_sector.py (Supplementary Table 43), analysis/unsure_group.py (Supplementary Table 44).